

When Do Standard Sector Taxonomies Fail?

Welfare-Consistent Aggregation in Quantitative Trade Models

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Abstract

Quantitative trade models routinely aggregate detailed products into coarse sectors. This paper shows that the choice of partition is not neutral: standard taxonomies group sectors by what they produce, while welfare analysis often requires grouping them by how shocks affect welfare. We formalize sector aggregation as a local welfare-preserving approximation problem and construct welfare-consistent partitions from sector-level welfare-exposure vectors. Using WIOD and high-resolution FIGARO-E3 benchmarks, we compare these partitions with standard taxonomies across shock families, compression regimes, and welfare objects. The results are conditional rather than universal. Taxonomies perform well for broad shocks, mild compression, and events whose incidence is defined on official product, service, or energy categories. Welfare-consistent partitions add value when compression is tight, when the training menu spans response directions relevant for targeted shocks, and—for events with sharply concentrated incidence—when the partition can be retrained on the event-specific geometry. Full-GE diagnostics show that changing the welfare object changes the relevant aggregation target. The findings imply that the key aggregation question is not only how many sectors to keep, but which welfare-relevant responses those sectors are meant to preserve.

1 Introduction

Quantitative trade models compress product-level and country-level heterogeneity into a manageable number of sectors. This compression is often treated as a computational necessity: fine-sector models can be too costly for large counterfactual grids, robustness exercises, estimation loops, or full-GE extensions. But computation is not the only reason researchers aggregate. Coarse sector systems also make results comparable across datasets and policy institutions, make policy incidence easier to communicate, and are often the only sectoral resolution available in applied settings. The practical question is not whether researchers should always use the finest available benchmark. It is whether the coarse model used for repeated analysis, comparison, or communication preserves the welfare responses relevant for the policy question. Once trade models incorporate heterogeneous trade elasticities, imported-input dependence, and production networks, the way sectors are grouped can

become a first-order determinant of welfare and policy inference. Two models with the same number of sectors may therefore deliver different counterfactuals not because one is “more disaggregated” than the other, but because one groups sectors along welfare-relevant margins while the other does not.

A large literature has already established that sectoral detail matters for gains from trade and for quantitative trade-policy analysis. In multi-sector models with intermediate inputs, sectoral heterogeneity and input-output linkages are central for quantifying the welfare effects of tariff changes (Caliendo and Parro, 2015). In settings with sectoral comparative advantage, one-sector sufficient-statistic formulas understate the gains from trade when there is meaningful cross-sector dispersion in trade patterns (Levchenko and Zhang, 2014). More recent work shows that the role of disaggregation depends not only on sector shares but also on how sectoral trade elasticities are estimated and pooled: re-estimating elasticities at different levels of aggregation can change the comparative welfare effect of disaggregation (Giri, Yi, and Yilmazkuday, 2021; Bolatto and Moramarco, 2023). This literature shows that sectoral heterogeneity matters. It leaves open the applied question of which coarse sector partitions best preserve the welfare-relevant structure of a fine benchmark economy.

That question is closely connected to aggregation theory. Anderson (2009) argues that trade-policy aggregation should preserve real income or real activity if it is to be meaningful for empirical counterfactuals. Redding and Weinstein (2017) show that the passage from micro to macro trade patterns introduces non-trivial composition terms, so aggregate objects need not inherit the structure of their disaggregated counterparts. Redding and Weinstein (2019) make this point explicitly for the gravity equation: because gravity is log-linear while aggregate trade is constructed by summing levels, aggregation introduces composition terms that affect aggregate gravity relationships. In related work, Breinlich, Novy, and Santos Silva (2024) show that aggregate gravity coefficients can often be interpreted as weighted averages of sector-level elasticities under specific conditions. These papers sharpen the issue studied here. Aggregation is not just a loss of detail; different aggregation rules preserve different economic objects. Preserving an average gravity coefficient is not the same as preserving local welfare responses to policy shocks.

The issue becomes sharper still in production-network environments. Once shocks propagate through domestic and international input-output linkages, the welfare relevance of a sector depends not only on its own expenditure share or trade elasticity, but also on its position in the network. Baqaee and Farhi (2024) develop a framework in which barriers and distortions diffuse through international production networks, while Bosker and Westbrock (2024) show that the gains from trade admit a network interpretation based on upstream and downstream propagation. These insights suggest that aggregation risk should be greatest in precisely those environments where shocks diffuse broadly through the economy. In the empirical work below, broad uniform shocks and targeted upstream- or downstream-intensive shocks emerge as useful stress tests for aggregation.

This paper studies sector aggregation as a *welfare-preserving approximation problem*. Starting from a high-resolution benchmark economy, we compute local welfare-exposure vectors for each

fine sector and ask how sectors should be grouped so as to preserve the welfare response of the benchmark. The resulting partition is a *welfare-consistent clustering* (WCC). This framing shifts the emphasis away from the number of sectors per se and toward the way sectors are grouped at a given model size. The central question is not only whether to keep more sectors, but whether the sectors that remain are grouped along welfare-relevant margins.

The economic distinction is between two notions of sector similarity. Standard taxonomies group sectors by production identity: technologies, product descriptions, and institutional classification rules. Welfare-consistent clustering groups sectors by counterfactual incidence: how a shock to a sector changes price indices, input costs, income incidence, and welfare across countries. These two notions can coincide for broad shocks or mild compression, but they need not coincide for targeted shocks whose incidence depends on trade exposure or network position. The paper therefore asks when production identity is a good proxy for counterfactual incidence, and when it is not.

The paper also distinguishes between the shocks used to *train* an aggregation and the shocks used to *evaluate* it. A partition that is optimal for one family of shocks need not be optimal for another. Likewise, a partition learned in a model with input-output linkages may or may not generalize to a regime in which those linkages are switched off. This motivates a cross-evaluation design in which the regime used to construct labels and the regime used to evaluate them are allowed to differ. The comparison is useful because it reveals whether changes in performance are driven by genuine economics or by regime-specific relabeling.

This perspective also clarifies how the paper relates to applied policy work. In climate-policy CGE models, [Caron \(2012\)](#) shows that sectoral aggregation can materially affect carbon leakage and the measured efficiency of border adjustments. The broader lesson is that aggregation may change not only the magnitude of policy effects, but also the ranking of competing policies. The present paper brings that insight into a modern quantitative-trade setting built on high-resolution global MRIO data.

Empirically, the paper proceeds in two steps. The first is a canonical Caliendo–Parro style exercise using WIOD data. It shows, in a familiar environment, that the aggregation problem is not specific to a particular high-resolution MRIO dataset. The second step is the main quantitative laboratory, based on the 2015 FIGARO-E3 product-by-product benchmark. The analysis compares welfare-consistent partitions with canonical FIGARO-64, a 73-group taxonomy-like baseline, random partitions, and balance-constrained variants. The baseline object is fixed-wage local price-index welfare, which isolates the aggregation margin in a transparent counterfactual environment. Full-GE diagnostics then ask what changes when the welfare object becomes real income and wage responses enter the approximation target.

The empirical findings are organized around four results. First, standard taxonomies are not inherently weak: they perform well for broad uniform shocks, mild compression, and events whose incidence is already aligned with official product or energy categories. Second, welfare-consistent clustering adds value when counterfactual incidence is not well summarized by production identity, but only when the training menu spans the relevant welfare-response directions. Third, compression

changes the ranking: as the number of retained sectors falls, response-based grouping becomes more valuable because taxonomy-like aggregation discards more welfare-relevant variation. Fourth, changing the welfare object changes the right aggregation target. Full-GE diagnostics are therefore not treated as a final ranking of taxonomies versus WCC; they test whether the appropriate aggregation changes with the welfare object. Taken together, the evidence suggests that the economically relevant question is not whether standard taxonomies are always too coarse, but when they are adequate, what dimension of policy transmission one wants to preserve, and when welfare-aligned partitions add value.

Section 6.3 explains how the same diagnostic logic can be used *ex post* for completed events whose incidence maps can be specified from observed policy or market information. In that setting, event-specific WCC is best read as a way to amortize fine-model response information into a coarse partition for repeated post-event analysis, not as an out-of-sample forecasting tool.

Related literature and contribution. This paper sits at the intersection of five literatures.

First, it contributes to the quantitative-trade literature on sectoral heterogeneity. The central lesson of [Caliendo and Parro \(2015\)](#), [Levchenko and Zhang \(2014\)](#), and [Giri, Yi, and Yilmazkuday \(2021\)](#) is that disaggregation matters for welfare. The contribution relative to that literature is not to revisit the one-sector-versus-multi-sector comparison, but to ask which coarse partitions best preserve the welfare response of a given high-resolution benchmark.

Second, the paper connects to the aggregation-theory literature. [Anderson \(2009\)](#) studies policy aggregation that preserves real income or real activity, while [Redding and Weinstein \(2017\)](#), [Redding and Weinstein \(2019\)](#), and [Breinlich, Novy, and Santos Silva \(2024\)](#) show that aggregation changes the mapping from micro parameters to aggregate objects, including the gravity equation. The contribution here is methodological rather than theoretical: we adopt the perspective that different aggregation rules preserve different objects, and provide an applied diagnostic procedure for choosing partitions that best preserve local welfare responses in quantitative-trade models. We do not claim a formal extension of these aggregation-theory results to welfare counterfactuals; the connection is conceptual.

Third, the paper contributes to the production-network literature. [Baqae and Farhi \(2024\)](#) and [Bosker and Westbrock \(2024\)](#) imply that the welfare relevance of a sector depends on its network exposure, not just on its own expenditure share. The paper implements this idea by clustering sectors using their welfare-exposure vectors and by treating targeted upstream and downstream shocks as empirical stress tests. The results also show why these tests must be interpreted through the lens of target alignment: network-shaped shocks do not automatically favor WCC unless the training menu spans the relevant response directions.

Fourth, the paper contributes to applied work on aggregation bias in policy analysis. Relative to [Caron \(2012\)](#), the novelty here is to compare taxonomy-based and welfare-aligned aggregations inside the same MRIO family and to do so with an explicit separation between training and evaluation menus. This design makes it possible to ask not only whether aggregation matters, but why.

Fifth, the paper connects to the broader granular-heterogeneity tradition in quantitative spatial

and trade economics. [Dingel and Tintelnot \(2026\)](#) show that quantitative spatial models can miscalibrate in granular settings, where the idiosyncratic component of individual choices dominates the moments used to fit aggregate predictions. We ask a related but distinct question for sector aggregation in quantitative-trade models: when does within-sector heterogeneity in welfare-response geometry break standard sector-aggregation approximations, and what coarse partition best preserves the heterogeneity that matters for welfare? The connection is in spirit rather than in formalism: both lines of work ask when standard fitting or aggregation practices mask economically important within-cell heterogeneity, while the specific failure modes—granular sampling variance versus coarse aggregation bias—differ.

The rest of the paper proceeds as follows. Section 2 presents the model, the welfare object, the welfare-consistent aggregation problem, and the empirical implementation. Section 3 uses a canonical CP-WIOD exercise as a diagnostic lab for target alignment and compression. Section 4 presents the main FIGARO-E3 evidence, including the compression frontier, the canonical FIGARO-64 comparison, shock-family contrasts, event-incidence diagnostics, and mechanism/robustness checks. Section 5 studies how the comparison changes under full-GE welfare objects. Section 6 discusses external validity and practical guidance for applied work.

2 Framework and Empirical Design

2.1 Environment

There are N countries, indexed by $n, i \in \{1, \dots, N\}$, and S fine products or sectors, indexed by $s, r \in \{1, \dots, S\}$. In each country, a representative household has Cobb–Douglas preferences over sectoral composites:

$$U_n = \prod_{s=1}^S C_{ns}^{\alpha_{ns}}, \quad \sum_{s=1}^S \alpha_{ns} = 1. \quad (1)$$

Within each sector, goods are differentiated by origin, and trade follows a sectoral Fréchet environment as in [Caliendo and Parro \(2015\)](#). Let π_{nis} denote the benchmark expenditure share of destination country n on product s sourced from origin country i , and let $\theta_s > 0$ denote the sector-specific trade elasticity.

Production in country n , sector s , combines value added and intermediate inputs according to

$$c_{ns} = A_{ns}^{-1} w_n^{\beta_{ns}} \prod_{r=1}^S P_{nr}^{\gamma_{nrs}}, \quad \beta_{ns} + \sum_{r=1}^S \gamma_{nrs} = 1, \quad (2)$$

where A_{ns} is productivity, w_n is the wage, P_{nr} is the price index of input r in country n , β_{ns} is the value-added share, and γ_{nrs} is the input-output coefficient measuring the cost share of input r in sector s .

Trade wedges are denoted by τ_{nis} . Around a benchmark equilibrium, the baseline analysis solves

a fixed-wage local hat system. Let hats denote proportional changes. Then

$$\log \hat{c}_{ns} = -\log \hat{A}_{ns} + \beta_{ns} \log \hat{w}_n + \sum_{r=1}^S \gamma_{nrs} \log \hat{P}_{nr}, \quad (3)$$

$$\hat{P}_{ns}^{-\theta_s} = \sum_{i=1}^N \pi_{nis} (\hat{c}_{is} \hat{\tau}_{nis})^{-\theta_s}. \quad (4)$$

In the fixed-wage benchmark, $\hat{w}_n = 1$ unless otherwise stated. Equation (4) updates sectoral price indexes in hats using benchmark bilateral expenditure shares and sector-specific trade elasticities.

2.2 Welfare object and the no-IO ablation

The baseline welfare object is a fixed-wage local price-index measure. Let

$$\log \hat{W}_n = - \sum_{s=1}^S \alpha_{ns} \log \hat{P}_{ns}. \quad (5)$$

Equation (5) is transparent and computationally stable, but it should be interpreted as a local fixed-income cost-of-living welfare object rather than as full general-equilibrium welfare under endogenous wages.

To isolate the role of production networks, we also use a no-input-output ablation in which

$$\gamma_{nrs} = 0 \quad \forall (n, r, s), \quad \beta_{ns} = 1 \quad \forall (n, s). \quad (6)$$

This removes the propagation of shocks through intermediate-input linkages while leaving the trade block otherwise unchanged.

2.3 A full-general-equilibrium extension

The same framework can also be evaluated under a full general-equilibrium closure. In that extension, one solves for a vector of wage changes $\hat{w}$ such that country-level labor-market or expenditure conditions hold. Conditional on $\hat{w}$, the price block remains the same as in (3)–(4); the additional step is to solve the wage block jointly with output. In the CP-WIOD GE diagnostic, labor endowments are fixed, baseline trade imbalances are held fixed, tariff revenue enters importer final expenditure, and wages are iterated until the country trade-balance residual is zero. Given wages, the counterfactual bilateral shares are

$$\pi'_{nis} = \frac{\pi_{nis} (\hat{c}_{is} \hat{\tau}_{nis})^{-\theta_s}}{\sum_{\ell} \pi_{n\ell s} (\hat{c}_{\ell s} \hat{\tau}_{n\ell s})^{-\theta_s}}.$$

Let t'_{nis} denote the gross counterfactual tariff factor used for revenue accounting, define $q_{nis} = \pi'_{nis}/t'_{nis}$, and let $F_{ns} = \sum_i q_{nis}$. The implementation treats baseline value added as the fixed labor endowment, so $Y_n(\hat{w}) = \hat{w}_n VA_n$, where $VA_n = \sum_s VA_{ns}$. Let S_n be the CP2015 surplus convention: $S_n > 0$ is a trade surplus and reduces final expenditure. Sectoral expenditure solves the

linear system

$$E_{nr} = \alpha_{nr} \{Y_n(\hat{w}) - S_n\} + \sum_s \gamma_{nrs} \sum_m q_{mns} E_{ms} + \alpha_{nr} \sum_s (1 - F_{ns}) E_{ns}. \quad (7)$$

The second term is intermediate demand by producers in country n , and the last term rebates importer tariff revenue to final demand according to α_{nr} . For pure iceberg trade-cost shocks, $t'_{nis} = 1$, so this tariff-revenue term is zero. Wages are then updated until

$$R_n(\hat{w}) = \sum_s F_{ns} E_{ns} - \sum_s \sum_m q_{mns} E_{ms} + S_n = 0,$$

where the first term is net-of-tariff absorption by country n and the second is producer sales by country n . Reported nominal hats are normalized by a numeraire country $\bar{n}$, $\tilde{w}_n = \hat{w}_n / \hat{w}_{\bar{n}}$ and $\tilde{P}_{ns} = \hat{P}_{ns} / \hat{w}_{\bar{n}}$, leaving real-income changes invariant. Real-income welfare is then

$$\log \hat{W}_n^{GE} = \log \hat{w}_n - \sum_s \alpha_{ns} \log \hat{P}_{ns}.$$

Section 5 uses full-GE diagnostics to study a change in welfare object: from local price-index preservation to real-income preservation. The baseline fixed-wage object remains the cleanest environment for isolating the aggregation margin, while the GE diagnostics ask whether the same object-alignment logic carries over once wage incidence enters the target.

2.4 Aggregation as a welfare-preserving approximation problem

Let $\mathcal{M}^{\text{train}}$ denote a training menu of local policy shocks. For each fine sector s , define a welfare-exposure vector

$$x_s \in \mathbb{R}^D, \quad (8)$$

where x_s stacks country-level welfare responses across a set of training shocks that load on sector s . In the empirical implementation, these vectors are computed by finite differences around the benchmark. If the training menu contains $|\mathcal{M}^{\text{train}}|$ sector-indexed shocks and the model has N countries, the raw exposure stack has dimension $D = N \times |\mathcal{M}^{\text{train}}|$ before any family-level transformations. When several shock families are stacked together, each family block is weighted-z-scored across sectors using the clustering weights and then multiplied by a family scalar weight. This prevents a family from dominating the clustering objective solely because it is measured in larger units; Appendix diagnostics use alternative training menus to show how performance changes when response directions are removed.

A coarse sector partition is a mapping $g : \{1, \dots, S\} \rightarrow \{1, \dots, K\}$ with $K < S$. Let $\omega_s \geq 0$ denote a sector weight. The baseline uses global gross-output weights, $\omega_s = \sum_n X_{ns}$; final-demand

weights are available as a robustness option. The welfare-consistent clustering solves

$$\min_{\{g(s)\}_{s=1}^S} \sum_{k=1}^K \sum_{s:g(s)=k} \omega_s \|x_s - \bar{x}_k\|^2, \quad \bar{x}_k = \frac{\sum_{s:g(s)=k} \omega_s x_s}{\sum_{s:g(s)=k} \omega_s}. \quad (9)$$

Equation (9) groups sectors by similar welfare responses rather than by broad production codes alone.

Proposition 1 (Local welfare approximation). *Suppose that, for shocks in the training neighborhood, fine-economy welfare responses admit the first-order representation*

$$\Delta \log W_n(m) = \sum_{s=1}^S b_{ns}(m) \delta_s + o(\|\delta\|),$$

where $m \in \mathcal{M}^{\text{train}}$ indexes the policy experiment, δ_s is the sector-level shock, and $x_s = \{b_{ns}(m)\}_{n,m}$ stacks the corresponding welfare-exposure coefficients. Consider a K -sector approximation that represents all fine sectors in cluster k by a common exposure vector $\bar{x}_k$. If the training design is orthogonal across sectors, or equivalently if the sector-level shock covariance used to define expected error is diagonal with weights ω_s , then the partition that minimizes expected first-order squared welfare-response error over the training menu is the solution to (9).

Sketch. The first-order representation maps each fine sector into the welfare-exposure vector x_s . Replacing sector s by its cluster representative creates first-order approximation error $x_s - \bar{x}_{g(s)}$. Weighting by ω_s and summing across fine sectors gives

$$\sum_{s=1}^S \omega_s \|x_s - \bar{x}_{g(s)}\|^2,$$

which is minimized, for a fixed assignment g , by the weighted mean in (9). Minimizing over assignments therefore gives the K -sector partition that best preserves the training-menu welfare-response geometry in local mean squared error. $\square$

The proposition gives the clustering problem its economic interpretation and also clarifies its limitation. For arbitrary multi-sector shocks, the full first-order expected-error objective contains cross-sector covariance terms. Equation (9) is exact for orthogonal sector-by-sector training designs and is otherwise a diagonal-covariance approximation to that fuller objective.¹ The proposition treats ω_s as the diagonal entries of an arbitrary diagonal shock covariance; the gross-output weighting

¹The cross-sector contribution to expected first-order error equals the Frobenius inner product of the off-diagonal part of the shock covariance matrix with the partition's exposure-residual Gram matrix, and is bounded above by the product of their Frobenius norms. Two summary diagnostics of the relative gap are therefore the off-diagonal-to-diagonal Frobenius-mass ratio of the shock covariance and its condition number; small values of both, evaluated against the partition residual norm, indicate the diagonal approximation is tight. [TBD: Hong reports both for the baseline FIGARO-E3 training menu, with corresponding values for the upstream-bundle and productivity menus alongside the welfare-error tables.]

in the empirical implementation is an economically motivated emphasis on larger sectors rather than a literal calibration of shock variance. It justifies the partition-selection objective in exposure space; it does not imply that the resulting coarse accounting model mechanically minimizes finite-shock welfare error, because the evaluated model is constructed by aggregating flows and recomputing structural shares. A taxonomy-based partition treats production identity as a proxy for the welfare incidence of shocks. Welfare-consistent clustering instead treats the exposure vector x_s as the relevant local sufficient statistic. In-sample performance reflects how well a partition preserves the response directions in $\mathcal{M}^{\text{train}}$; out-of-sample performance depends on whether the evaluation shocks load on the same local welfare-response geometry.

2.5 Constructing a coarse counterfactual model

The aggregation exercise has two distinct steps. Equation (9) chooses a partition. Given any partition, the coarse model is then constructed by aggregating levels and recomputing the structural shares from those aggregated flows. This ensures that taxonomy-based partitions, welfare-consistent partitions, and random partitions are all evaluated through the same accounting map.

Let $M_{ks} = \mathbf{1}\{g(s) = k\}$ denote the fine-to-coarse mapping matrix. For any partition g , the coarse intermediate-use, final-demand, output, and value-added objects are

$$Z_{ink\ell}^G = \sum_{r=1}^S \sum_{s=1}^S M_{kr} M_{\ell s} Z_{inrs}, \quad F_{ink}^G = \sum_{r=1}^S M_{kr} F_{inr}, \quad (10)$$

$$X_{nk}^G = \sum_{s=1}^S M_{ks} X_{ns}, \quad VA_{nk}^G = \sum_{s=1}^S M_{ks} VA_{ns}. \quad (11)$$

The coarse benchmark shares are then recomputed from these coarse levels:

$$\pi_{nik}^G = \frac{\sum_{\ell} Z_{ink\ell}^G + F_{ink}^G}{\sum_j \left(\sum_{\ell} Z_{jnkl}^G + F_{jnk}^G \right)}, \quad (12)$$

$$\alpha_{nk}^G = \frac{\sum_i F_{ink}^G}{\sum_h \sum_i F_{inh}^G}, \quad (13)$$

$$\gamma_{nhk}^G = \frac{\sum_i Z_{inhk}^G}{X_{nk}^G}, \quad \beta_{nk}^G = \frac{VA_{nk}^G}{X_{nk}^G}. \quad (14)$$

Thus the coarse model is not obtained by averaging primitive shares directly. It is obtained by first aggregating the underlying flows and then reconstructing the compact CP objects

$$(\pi^G, \gamma^G, \alpha^G, \beta^G, X^G, VA^G).$$

When sector-specific trade elasticities are used, the baseline pooling rule is an output-weighted harmonic mean,

$$\theta_k^G = \frac{\sum_s M_{ks} \Omega_s}{\sum_s M_{ks} \Omega_s / \theta_s}, \quad \Omega_s = \sum_n X_{ns}. \quad (15)$$

Table 1: WCC label types used in the empirical design

Label type	Training object	Evaluation object	Interpretation
Baseline WCC	Generic menu	Generic or target menu	Feasible response-based partition
Holdout WCC	Excludes target family	Target family	Generalization test
Family-oracle WCC	Evaluation family	Same family	In-sample family diagnostic
Event-oracle WCC	Event incidence map	Same event	Forward-looking upper bound; retrospective event-specific diagnostic

Notes: The table describes the interpretation of WCC labels, not different economic models. All partitions are evaluated by constructing a coarse model from the same underlying accounting map.

Uniform-elasticity runs are invariant to this pooling step. The implementation also permits arithmetic pooling as a robustness option. Trade-cost and productivity shocks are projected into the coarse model using the same accounting principle: shocks are averaged within coarse cells using the relevant baseline flow or output weights.

2.6 Training menus, evaluation menus, and cross-evaluation

A partition that is optimal for one family of shocks need not be optimal for another. We therefore distinguish between a *training menu* $\mathcal{M}^{\text{train}}$, used to construct the labels in (9), and an *evaluation menu* $\mathcal{M}^{\text{eval}}$, used to assess out-of-sample performance.

Table 1 summarizes the label types used throughout the empirical sections. This distinction is important because the same word “WCC” can refer either to a feasible partition trained on a generic policy menu or to an oracle diagnostic trained on the evaluation object itself.

We also distinguish between the regime used to construct labels and the regime used to evaluate them. Let $R \in \{\text{with-IO}, \text{no-IO}\}$ denote the regime used for clustering and let $R' \in \{\text{with-IO}, \text{no-IO}\}$ denote the regime used for evaluation. This yields a 2×2 cross-evaluation design:

$$(\text{labels regime } R) \times (\text{evaluation regime } R').$$

If regime-specific relabeling is unimportant, then a clustering trained in one regime should perform similarly in the other. If performance instead depends primarily on the alignment between the training target and the evaluation menu, then the main differences should arise across menus rather than across regimes.

2.7 Evaluation and decomposition

For a given evaluation menu, let $\Delta \log W_{pn}^{\text{fine}}$ denote the welfare effect of policy p in country n under the fine benchmark and let $\Delta \log W_{pn}^G$ denote the corresponding effect under aggregation G . We summarize approximation quality with two main metrics:

$$\text{MAE}(G) = \text{median}_{p,n} \left| \Delta \log W_{pn}^G - \Delta \log W_{pn}^{\text{fine}} \right|, \quad (16)$$

$$\text{Reversal}(G) = \text{fraction of policy pairs whose ranking changes under } G. \quad (17)$$

The median in MAE is taken over unweighted policy-country cells unless a table explicitly reports a normalized or event-specific relative error. The reversal statistic compares policy rankings within each country and then averages across countries; singleton shock-family rows therefore do not have a meaningful reversal rate. The implementation also computes country-level Spearman rank correlations and allows for tolerance thresholds when treating near ties.

Finally, we decompose total approximation error into a structural aggregation component and a theta-pooling component. Let $\Delta \log W^{\text{pool}}$ denote the welfare response of the *fine* model evaluated using pooled trade elasticities. Then

$$E^{\text{total}} = \Delta \log W^G - \Delta \log W^{\text{fine}}, \quad (18)$$

$$E^{\theta\text{-pool}} = \Delta \log W^{\text{pool}} - \Delta \log W^{\text{fine}}, \quad (19)$$

$$E^{\text{struct}} = \Delta \log W^G - \Delta \log W^{\text{pool}}. \quad (20)$$

In the uniform-elasticity benchmark, all sectors are assigned the same elasticity, $\theta_s = 4$, so the theta-pooling term is mechanically uninformative in the headline runs. The decomposition is nevertheless useful because it clarifies how results with sector-specific elasticities should be interpreted. We also report a partial heterogeneous-theta robustness exercise based on the ProTEE product-level estimates of [Fontagné, Guimbard, and Orefice \(2022\)](#), mapped from HS6 products through CN2015 and CPA2.1 into the FIGARO product space. Unmatched services and residual categories are filled with ICIO/TiVA-style sector averages or the transparent fallback value used in the uniform runs. That robustness check leaves the main ranking patterns largely unchanged, which suggests that structural aggregation remains the first-order margin.

2.8 Data and benchmarks

2.8.1 Two empirical layers

The empirical analysis has two layers. The first is a compact Caliendo–Parro style exercise built from WIOD. This exercise is intended to be small and transparent. It uses a familiar quantitative-trade environment to show that aggregation can change counterfactual welfare responses even before moving to the richer FIGARO-E3 product space. Its role is proof of concept rather than the main quantitative benchmark.

The second layer is the high-resolution FIGARO-E3 exercise, which serves as the main empirical laboratory of the paper. FIGARO-E3 provides a product-by-product global MRIO structure with enough sectoral detail to compare conventional taxonomy-based aggregation against welfare-consistent clustering across compression regimes, shock families, input-output regimes, and robustness exercises.

2.8.2 FIGARO-E3 benchmark

The empirical benchmark is built from the 2015 FIGARO-E3 product-by-product matrix. The benchmark contains 46 countries, including the rest-of-world aggregate, and 213 fine products. It offers a high-resolution global MRIO environment while remaining tied to the official FIGARO statistical family. This makes it especially well suited for studying how much welfare-relevant information is lost when one moves from a fine product classification to a coarser sectoral representation.

The paper works in product space rather than industry space because the policy objects of interest—trade-cost wedges, upstream-intensive shocks, and product-level productivity changes—are most naturally indexed by products. The main official comparison uses the canonical FIGARO-64 concordance. We also report a native 73-group taxonomy-like baseline obtained by collapsing fine FIGARO-E3 product codes through suffix stripping. This $K = 73$ comparator is useful because it provides a strong taxonomy-like baseline, but it should not be interpreted as a canonical official concordance.

2.8.3 Benchmark construction

We convert the FIGARO-E3 product-by-product MRIO into a compact benchmark containing bilateral expenditure shares, final-demand shares, input-output cost shares, value-added shares, gross output, and elasticity inputs. Appendix A describes the raw-matrix parsing, cleaning, zero-output regularization, and cost-share consistency checks. The main text uses the resulting compact arrays to keep the empirical discussion focused on the aggregation object rather than the engineering details of the raw matrix.

For the numerical exercises, the benchmark enforces column-side cost-share consistency. Gross output is measured as intermediate demand by column plus value added, so that

$$\beta_{ns} + \sum_r \gamma_{nrs} = 1$$

by construction. This is the identity required by the fixed-wage price block. The compact benchmark then computes

$$\pi_{nis} = \frac{\sum_t Z_{inst} + F_{ins}}{\sum_j (\sum_t Z_{jnst} + F_{jns})}, \quad (21)$$

$$\alpha_{ns} = \frac{\sum_i F_{ins}}{\sum_r \sum_i F_{inr}}, \quad (22)$$

$$\gamma_{nrs} = \frac{\sum_i Z_{inrs}}{X_{ns}}, \quad \beta_{ns} = \frac{VA_{ns}}{X_{ns}}. \quad (23)$$

The resulting benchmark stores only the compact arrays needed by the numerical core:

$$(\pi, \gamma, \alpha, \beta, X, VA).$$

At that point, the raw tensors can be set aside for numerical purposes.

3 CP-WIOD Diagnostic Lab

3.1 Canonical proof-of-concept and target alignment

Before turning to the high-resolution FIGARO-E3 evidence, we first run a compact exercise in a canonical Caliendo–Parro style environment using the 2014 WIOD release (Timmer et al., 2015). The purpose is to separate the conceptual point from the details of the FIGARO dataset. WIOD is much coarser than FIGARO-E3, but it has the advantage of being a familiar global input-output environment. Finding aggregation effects in this setting makes the subsequent FIGARO results easier to interpret as a high-resolution extension rather than as a dataset-specific artifact.

The WIOD benchmark contains 44 countries, including the rest of the world, and 56 sectors. We treat the 56-sector table as the fine benchmark and assign CP-style heterogeneous trade elasticities by mapping the tradable-sector estimates in Caliendo and Parro (2015) to the WIOD sector grid, with a transparent fallback for services and nontradables not covered by the original estimates. We then compare two $K = 20$ partitions: an ISIC-section taxonomy and a welfare-consistent clustering trained on the same compact policy menu used for evaluation. The purpose is not to establish the final quantitative magnitudes, but to show whether the aggregation problem appears in a standard CP-style setting before moving to the richer FIGARO-E3 product space.

Appendix Table B.1 reports the simplest CP-WIOD proof-of-concept. It establishes two points. First, in this canonical setting the two economically motivated partitions are close overall and both dominate a random partition. This is a useful benchmark: the paper is not claiming that standard taxonomies fail mechanically. Second, the shock-family comparison already points in the same direction as the main FIGARO results. The ISIC-section taxonomy is more accurate for the uniform trade-cost shock, while welfare-consistent clustering is substantially more accurate for the upstream-intensive shock. Thus even in WIOD, aggregation quality is better understood as shock-specific target alignment than as a universal ranking of taxonomies versus clustering.

The economics of this contrast is intuitive. A uniform trade-cost shock moves many sectors in the same direction and loads heavily on broad features such as tradability, import intensity, and average expenditure structure. An ISIC-section taxonomy can preserve much of that common movement. An upstream-intensive shock, by contrast, is an incidence shock: what matters is not only which sectors are directly hit, but also which sectors sell disproportionately to other producers and how those input-cost changes propagate through country-sector production networks. A partition that groups sectors by response similarity can therefore preserve margins that are not visible in sector names alone.

To ask whether the welfare-consistent advantage is specific to upstream shocks, we add two holdout diagnostics to the WIOD exercise. The first is a downstream-intensive trade-cost shock that loads on the sectors most directly absorbed by final demand. In the holdout version, the downstream shock is excluded from the WCC training menu and used only for evaluation. Welfare-consistent

Table 2: CP-WIOD target alignment at $K = 20$: WCC/taxonomy error ratios

WCC training variant	All	Uniform	Upstream	Downstream	Productivity
Target-aligned	0.943	0.712	0.452	0.195	0.956
Productivity holdout	1.032	0.870	0.439	0.231	0.920
Network holdout	0.880	0.592	0.864	0.545	0.738
Network and productivity holdout	1.090	0.988	0.944	0.647	1.223

Notes: Entries below one favor welfare-consistent clustering. The evaluation menu contains 117 shocks: 56 sector-global trade-cost shocks, one uniform trade-cost shock, targeted bilateral trade-cost shocks, upstream and downstream bundle shocks, and 56 USA sector-productivity shocks. “Network holdout” excludes upstream and downstream bundle shocks from WCC training. “Network and productivity holdout” excludes upstream, downstream, and productivity shocks from WCC training. Single-sector trade-cost shocks remain in the WCC training menu in every row, so the All column should be read as in-distribution for that shock family.

clustering remains more accurate than the ISIC-section taxonomy, with a WCC/taxonomy error ratio of 0.736. When the downstream shock is included in the training menu, the ratio falls further to 0.201. The second diagnostic initially considers a narrow single-sector productivity shock. That single-shock holdout favors WCC, but the broader productivity diagnostic below qualifies the interpretation. Once the evaluation menu contains productivity shocks for all 56 WIOD sectors in a large anchor country, WCC need not generalize from unrelated training directions. Its performance depends on whether the training menu contains response directions that span the productivity family. These exercises suggest that the relevant distinction is not upstream versus downstream per se, or trade-cost versus productivity shocks, but whether the policy incidence is better summarized by welfare-response geometry than by broad sector identity.

Table 2 makes this target-alignment point explicit. At $K = 20$, WCC is strong for downstream and upstream bundles when the training menu contains relevant network directions. But when both network and productivity shocks are removed from the training menu, WCC deteriorates on the pooled all-shock metric and underperforms the taxonomy for the expanded productivity family. Conversely, when productivity shocks remain in the training menu while network bundles are held out, WCC is substantially more accurate for productivity shocks. Thus WCC is not a rule that automatically wins for every network or productivity shock. It is a target-specific aggregation rule whose performance depends on whether the training menu spans the welfare-response geometry of the evaluation shock.

The uniform column also illustrates why the paper avoids a universal statement that broad shocks always favor taxonomy. Appendix Table B.1 uses a simpler proof-of-concept menu and finds that the ISIC-section taxonomy is more accurate for the baseline uniform shock. Table 2 instead uses the expanded 117-shock target-alignment design and the WCC labels trained on that richer menu; under that design the uniform ratio falls below one. Thus broad uniform shocks are best interpreted as cases in which taxonomy is especially competitive, not as cases in which taxonomy must always dominate. In the canonical FIGARO-64 comparison below, the broad uniform shock favors taxonomy strongly.

Appendix Table B.2 pushes the same point one step further by comparing restricted or holdout

WCC labels, target-aligned WCC labels, and family-oracle WCC labels. Appendix Table B.4 then reports representative CP-WIOD clusters. These diagnostics support the main lesson in Table 2: WCC is useful when the training menu spans the relevant response directions richly enough, not simply when the evaluation shock is labeled as a network or productivity shock.

3.2 Compression frontier

The same target-alignment logic also depends on compression. A mixed-target K -frontier shows that the family-balanced median WCC/taxonomy ratio is not monotone in K or invariant across training menus. The best-performing variant changes with K , and the gap between target-aligned and holdout WCC is largest when the partition has enough flexibility to represent the relevant margins. Figure 1 makes this visible by plotting WCC/taxonomy error ratios across K and training variants. For K values other than the native $K = 20$ ISIC-section partition, the taxonomy frontier is constructed mechanically from the native taxonomy: when the native taxonomy is too fine, native labels are coarsened into K bins; when it is too coarse, the largest bins are split until the requested K is reached. This is a taxonomy-like compression diagnostic rather than an alternative official concordance. A separate trade-cost frontier with a balance cap, $\lceil 3S/K \rceil$, reduces the largest WCC cluster substantially—for example from 32 to 21 at $K = 8$ and from 32 to 11 at $K = 16$. The downstream advantage remains in place under this cap. The upstream pattern remains non-monotone, however, suggesting that the intermediate- K reversals are not merely a giant-cluster artifact.

4 High-Resolution FIGARO-E3 Evidence

4.1 Benchmark choices and policy menus

The FIGARO exercise has three benchmark choices that guide the interpretation of the results.

First, the $K = 73$ taxonomy-like comparator is a heuristic 73-group partition rather than a canonical FIGARO concordance. It is constructed by dropping variant suffixes from fine product codes. This can create a strong baseline when the fine products are already defined as narrow variants inside broad sectors. The exact canonical comparison is therefore the FIGARO-64 exercise at $K = 64$. Interpretations should distinguish clearly between the exact canonical comparison at $K = 64$, the coerced official baselines at lower K , and the native 73-group taxonomy-like benchmark at $K = 73$.

Second, the baseline benchmark sets

$$\theta_s = 4 \quad \forall s.$$

This uniform-elasticity choice is useful for isolating structural aggregation effects, but it renders the theta-pooling component of the decomposition mechanically uninformative in baseline runs. We therefore supplement the baseline with a partial heterogeneous-theta robustness exercise built

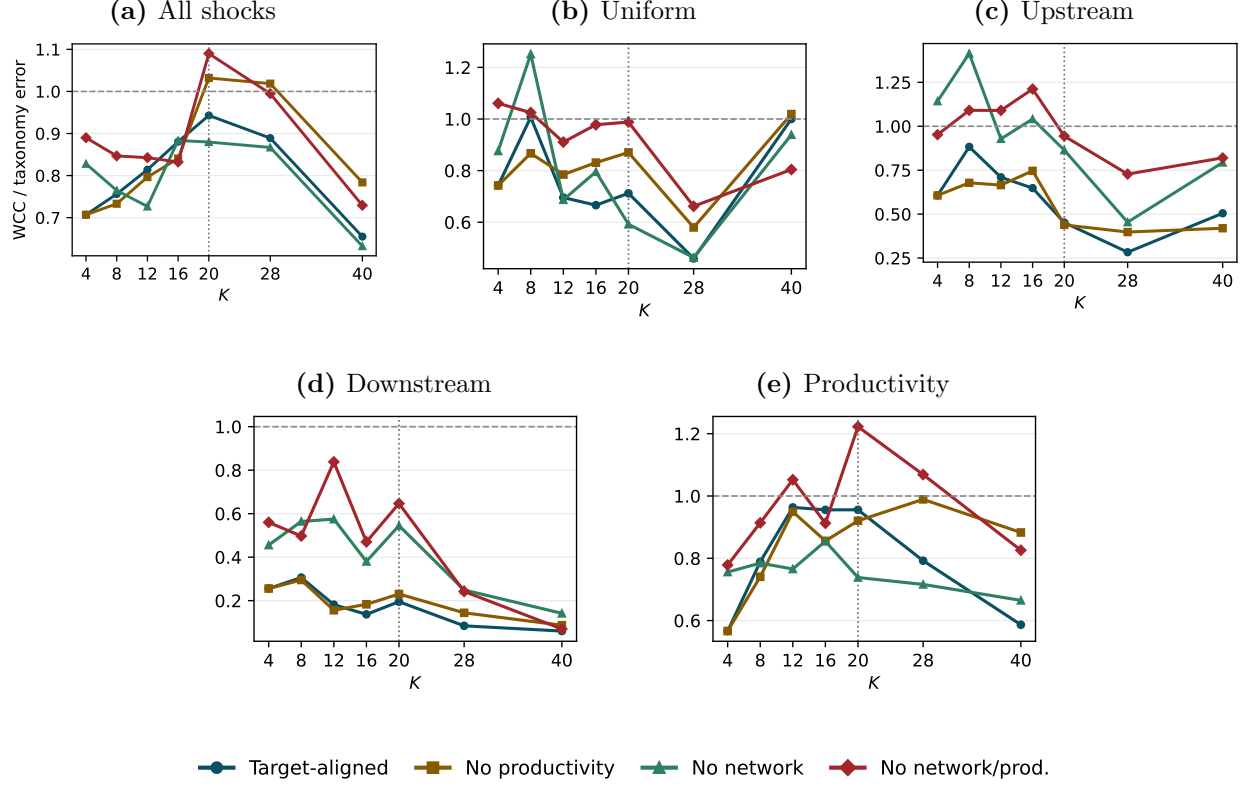


Figure 1: CP-WIOD K -frontier: compression and target alignment

Notes: The figure reports WCC/taxonomy welfare-error ratios from the CP-WIOD mixed-target frontier. Values below one favor welfare-consistent clustering. The vertical dotted line marks the native ISIC-section partition at $K = 20$; the taxonomy comparator is exact at that value and is the coerced taxonomy frontier at other values of K . WCC labels and evaluation are both computed in the with-IO regime. Training variants exclude the indicated shock families from the WCC training menu while retaining them in the evaluation menu.

from ProTEE HS6 elasticities (Fontagné, Guimbard, and Orefice, 2022) mapped into the FIGARO-E3 product classification. Goods are mapped through the CN2015–CPA2.1 bridge; services and unmatched residual categories use ICIO/TiVA-style sector fallbacks before elasticities are pooled with the harmonic rule described above.

Third, the main welfare object is the fixed-wage local price-index measure in equation (5). This choice keeps the price block transparent and makes the aggregation exercise easy to interpret. Section 5 then reports full-GE object-change diagnostics to show how the comparison changes once wage responses enter the welfare object.

The FIGARO comparisons should therefore be read in four layers. The main canonical comparison is exact FIGARO-64 versus $K = 64$ WCC. The compression frontier uses coerced FIGARO-64-style taxonomy baselines at $K = 32$ and $K = 48$ to vary the number of groups. The $K = 73$ exercise is a native suffix-based taxonomy-like comparison, not an official concordance. Balance constraints, random partitions, and heterogeneous-theta variants are robustness exercises around these main

comparisons.

4.1.1 Policy menus and empirical variation

The empirical implementation organizes shocks into explicit policy menus. The evaluation menu includes:

1. sector-global trade-cost shocks,
2. a broad uniform trade-cost shock,
3. importer-sector trade-cost shocks,
4. sector-country productivity shocks,
5. upstream-intensive trade-cost shocks,
6. bilateral importer-full-sector trade-cost shocks.

The training menu used to construct welfare-consistent partitions can differ from the evaluation menu. This distinction is central to the empirical design because it allows the paper to ask whether weaker aggregation performance reflects a limitation of the partition itself or simple misalignment between the clustering target and the policy menu.

We also report cross-evaluation across with-IO and no-IO regimes. This is useful because it distinguishes between genuine economic differences and regime-specific relabeling. These diagnostics summarize the comparison through decomposition terms and policy-level welfare errors, which remain directly comparable as elasticity and GE specifications are refined.

4.2 Compression and canonical FIGARO-64 comparison

The mechanism underlying the results is not that one partition is universally more detailed or more sophisticated than another. It is that different partitions preserve different economic margins. Taxonomies preserve production identity and broad technological similarity. Welfare-consistent partitions preserve local counterfactual incidence: the pattern of welfare responses induced by trade exposure, intermediate-input use, final-demand absorption, elasticities, and network propagation. The empirical question is therefore when these two concepts of similarity coincide. They are more likely to coincide for common shocks and mild compression, and more likely to diverge when policy incidence is targeted or when compression forces economically distinct fine sectors into the same coarse cell.

The results are economically interpretable after two mechanical concerns are addressed. Exact- K repair ensures that welfare-consistent clustering uses the requested number of groups, and full-coverage training prevents the K -frontier from being mechanically flat. Balance-constrained robustness checks further show that the main patterns do not rely on extremely oversized clusters. The relevant question is therefore not whether the results reflect mechanical clustering artifacts, but

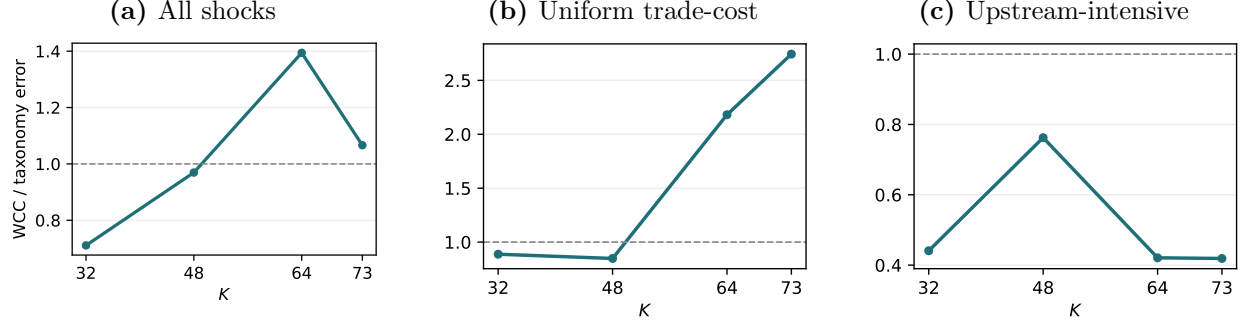


Figure 2: FIGARO fixed-wage K -frontier: WCC relative to taxonomy

Notes: The figure reports WCC/taxonomy welfare-error ratios from the FIGARO fixed-wage local price-index frontier. Values below one favor WCC. For $K = 32$ and $K = 48$, the taxonomy comparator is a coerced FIGARO-64 partition; at $K = 64$, it is exact canonical FIGARO-64; at $K = 73$, it is the native taxonomy-like suffix baseline. This frontier uses the uniform-elasticity full-coverage WCC labels; Appendix Table C.3 reports the balance-constrained robustness at $K = 64$ and $K = 73$.

which aggregation rule preserves fine-economy welfare responses more effectively once the number of coarse sectors is fixed.

The connection to Redding and Weinstein (2019) is useful for interpreting these patterns. Their aggregation correction explains why aggregate gravity objects need not reproduce the log-linear relationships that hold at more disaggregated levels. The present exercise asks what happens after such aggregated objects are used inside a counterfactual model. If fine sectors inside a coarse group differ in trade exposure, elasticity, or input-output position, the aggregate sector does not generally reproduce the fine economy’s price-index and welfare response to a shock. Counterfactual aggregation error can therefore be viewed as the welfare-response analogue of a gravity aggregation error.

The first pattern is that aggregation performance depends on the strength of compression, and this survives the balance-constrained robustness exercise. Low- K FIGARO frontier diagnostics indicate that welfare-consistent clustering can outperform both the suffix-based heuristic baseline and coerced FIGARO-64-style baselines when compression is tight. Under milder compression, however, taxonomy-based partitions are stronger. In the exact $K = 64$ comparison, canonical FIGARO-64 continues to outperform welfare-consistent clustering on the headline welfare-error row even after imposing a cap on the largest welfare-consistent cluster, which reduces that cluster from 91 sectors to 54. At $K = 73$, by contrast, balance-constrained welfare-consistent clustering slightly improves on the native 73-group taxonomy-like baseline in welfare magnitude, although not in ranking reversals. A natural interpretation is that, when the number of retained sectors is still relatively large, official taxonomies already preserve many of the broad technological and demand-side similarities that matter for average welfare responses. When compression becomes more binding, however, grouping sectors by welfare-response similarity becomes more valuable than grouping them by sector identity alone.

The labels themselves also show why the comparison is economic rather than purely algorithmic.

Appendix Table C.1 reports selected clusters from the balance-constrained $K = 64$ FIGARO WCC partition used in the heterogeneous-theta comparison. Some clusters remain close to familiar broad sectors, such as construction or real estate. Others cross canonical FIGARO-64 boundaries when sectors have similar response profiles. These are not natural product-code categories, but they are interpretable as response-based incidence groups with similar final-demand or network exposure.

The second pattern is shock-specific, but the distinction is not mechanical. Broad uniform trade-cost shocks often favor taxonomy-based partitions, while targeted network-sensitive shocks can favor welfare-consistent clustering. The extended CP-WIOD diagnostics show why the qualification matters: a network-shaped evaluation shock is not sufficient by itself. WCC performs well when the training menu spans the welfare-response directions relevant for the evaluation shock, and it can deteriorate when those directions are omitted. Uniform shocks are dominated by common-movement components such as tradability, import intensity, and broad expenditure structure, all of which standard taxonomies can preserve reasonably well. By contrast, upstream- and downstream-intensive shocks depend more heavily on production-network position and propagation, so an aggregation rule that groups sectors by response similarity can better preserve the relevant margins when the training objective contains those margins. In that sense, taxonomy and welfare-consistent clustering are not preserving the same object: taxonomy appears stronger for broad common-movement shocks, while welfare-consistent clustering appears stronger for targeted shocks whose incidence is well represented in the training menu.

Table 4 summarizes this pattern using WCC/taxonomy welfare-error ratios. Values below one favor welfare-consistent clustering, while values above one favor the taxonomy comparator. The ratio is above one for broad uniform shocks in the baseline FIGARO comparisons. It is below one for upstream-intensive shocks in both datasets, and it remains below one for the WIOD downstream holdout diagnostic. The single-productivity holdout row shows that WCC can perform well outside trade-cost shocks, but the expanded productivity diagnostic shows that this should be interpreted as evidence for target alignment rather than automatic cross-family generalization. The single-sector trade-cost rows are more mixed: the WCC win share is close to one half in WIOD and below one half in the FIGARO $K = 64$ comparisons. Thus the evidence does not support the stronger claim that any individual upstream sector automatically favors WCC. A more cautious interpretation is that WCC has a systematic advantage for targeted shock families whose welfare incidence is shaped by response margins represented in the training menu, while standard taxonomies are strong for broad common shocks.

Appendix Table B.3 checks that this CP-WIOD target-alignment result is not driven by one arbitrary family-weight choice in the WCC objective. Equal family weights and productivity-heavy weights preserve the qualitative pattern, while turning off family-level normalization weakens only the pooled all-shock row.

Appendix Table B.5 studies shock-size sensitivity in the CP-WIOD environment by holding the baseline WCC labels fixed and scaling the evaluation shocks. Absolute errors scale approximately proportionally with shock size, while family-specific ratios move with scale. This is useful as a

robustness check, but it is not a separate main result: it shows that a local response-based partition can remain informative away from its training neighborhood, while its relative performance remains object- and family-specific.

The third pattern is that regime-specific relabeling adds little incremental value in the baseline cross-evaluation design. Appendix Table C.2 reports the compact comparison. Holding the with-IO evaluation regime fixed, the WCC error is similar whether labels are trained with input-output linkages or in the no-IO ablation. Turning off input-output linkages sharply reduces aggregation error, especially for welfare-consistent clustering, but the gains from relabeling across regimes are small relative to the change in propagation environment. The most natural interpretation is that network propagation amplifies structural aggregation errors, so mistakes matter more when propagation is active.

Finally, the heterogeneous-theta robustness checks suggest that elasticity pooling is not the dominant explanation for the baseline results. Partial ProTEE-based heterogeneity leaves the main ranking at $K = 64$ in place, and the theta-pooling component remains small on the headline “all” row relative to structural aggregation error. The main exception is the broad uniform shock family, where heterogeneous theta begins to matter quantitatively. The balance-constrained robustness results sharpen this interpretation: even after materially reducing the largest welfare-consistent cluster, the taxonomy advantage under mild compression and the welfare-consistent advantage for upstream-intensive shocks remain in place. This is consistent with the view that the first-order issue in the present environment is how sectors are grouped, while elasticity heterogeneity acts as a secondary modifier of aggregation error.

4.3 Shock-family evidence

Table 3 is the canonical high-resolution comparison. Panel A reports the exact $K = 64$ comparison in the partial heterogeneous-theta specification with balance-constrained welfare-consistent clustering. Canonical FIGARO-64 has the smallest headline welfare-error row, while WCC remains much more accurate than the random benchmark. Panel B shows why the headline row is not the whole story: taxonomy is much stronger for the broad uniform shock, while WCC is much stronger for the upstream-intensive shock. Appendix Tables C.3–C.4 show that this contrast is not driven by an oversized residual cluster.

4.4 Event-incidence diagnostics: when taxonomies are the right object

The real-world event diagnostics show the other side of object alignment. They are not tests of whether a baseline WCC partition always beats taxonomy. They ask whether the event’s incidence map is already organized in the language of inherited sectors. When it is, a taxonomy-based partition can be the right aggregation object.

We consider four incidence maps. The USTR Section 301 diagnostic maps the official China tariff lists to FIGARO sectors and evaluates the implied U.S. import-cost shock from China (USTR, 2026). The Brexit diagnostic is a UK–EU services exercise: it follows the goods-versus-services discipline

Table 3: Canonical FIGARO-64 comparison and shock-family contrast

<i>Panel A. Exact $K = 64$ comparison</i>					
Aggregation	Effective K	Largest cluster	Median error	Relative error	Reversal rate
Noncanonical taxonomy-like $K = 64$ heuristic	64	22	3.28×10^{-6}	2.34%	0.887%
Canonical FIGARO-64	64	22	3.08×10^{-6}	2.20%	0.931%
Welfare-consistent clustering	64	54	3.73×10^{-6}	2.67%	1.010%
Random benchmark	64	6	9.60×10^{-6}	6.86%	1.738%
<i>Panel B. Shock-family contrast</i>					
Shock family	Canonical error	Canonical rel.	WCC error	WCC rel.	WCC/tax.
All shocks	3.08×10^{-6}	2.20%	3.73×10^{-6}	2.67%	1.210
Uniform trade-cost shock	4.25×10^{-5}	0.44%	1.03×10^{-4}	1.07%	2.432
Upstream-intensive trade-cost shock	2.31×10^{-5}	3.60%	7.98×10^{-6}	1.24%	0.346

Notes: Panel A reports fixed-wage local price-index welfare errors for the partial heterogeneous- θ FIGARO specification. Relative error divides the median absolute aggregation error by the median absolute fine-model response in the corresponding row or shock family. The noncanonical taxonomy-like row is a heuristic $K = 64$ repair and is not the main official comparator. Panel B reports median absolute welfare errors by shock family under the same specification. The canonical comparator is exact FIGARO-64; WCC uses the balance-constrained $K = 64$ partition. The canonical headline all-shock error is 3.08×10^{-6} .

in the Brexit literature but uses ONS service-type data to make finance, insurance, ICT, transport, professional services, technical services, travel, and related categories move differently ([Office for National Statistics, 2026](#)). The Russia–EU and Hormuz-style diagnostics are energy-incidence maps: EU import costs on Russian coal, oil and gas, and refined energy sectors, and import costs on Gulf-origin oil, gas, and refined-energy exports ([Eurostat, 2022](#); [European Commission, 2026](#); [U.S. Energy Information Administration, 2025](#)). None of these rows should be read as a realized welfare estimate. The historically motivated event map is the input; the question is which $K = 64$ partition best preserves the fine FIGARO welfare vector under that map.

Table 5 reports two main patterns. First, product-list and service-category events are taxonomy-friendly. Section 301 is written in official product language, and the Brexit services map remains close to recognizable official service categories even after the ONS exposure adjustment. In both cases, event-oracle WCC improves on the baseline WCC labels but remains less accurate than canonical FIGARO-64. Second, the energy events are concentrated in official energy sectors, so baseline WCC can perform badly when it dilutes the event support across clusters that were trained for other policy menus. In those same energy rows, however, event-oracle WCC becomes substantially more accurate than FIGARO-64, with ratios around 0.20–0.22 for Russia–EU energy and 0.26–0.31 for Hormuz-style energy.

The event-oracle rows should therefore be read in two registers. For forward-looking analysis, they are oracle benchmarks rather than out-of-sample victory claims: projecting an event’s incidence ahead of time is precisely the hard part. For retrospective analysis of a completed event, they are event-specific partitions that can be implemented ex post once the researcher specifies the incidence map, fine benchmark, and welfare object. Section 6.3 returns to this distinction. Either way, the table makes the same point as the rest of the paper: WCC is valuable because it diagnoses whether the policy object requires a different response geometry than the inherited classification system.

Table 4: Shock-family performance ratios: WCC relative to taxonomy

Dataset and run	Shock family	WCC/tax. ratio	WCC win share
WIOD baseline	Uniform trade-cost	1.203	0.00
WIOD baseline	Upstream-intensive	0.489	1.00
WIOD downstream holdout	Downstream-intensive	0.736	1.00
WIOD downstream target-aligned	Downstream-intensive	0.201	1.00
WIOD single-productivity holdout	Productivity shock	0.311	1.00
WIOD no network/prod. training	Productivity shocks	1.223	–
FIGARO $K = 64$, uniform θ	Uniform trade-cost	2.181	0.00
FIGARO $K = 64$, uniform θ	Upstream-intensive	0.390	1.00
FIGARO $K = 64$, het. θ	Uniform trade-cost	2.432	0.00
FIGARO $K = 64$, het. θ	Upstream-intensive	0.348	1.00
WIOD baseline	Single-sector trade-cost shocks	1.015	0.50
FIGARO $K = 64$, uniform θ	Single-sector trade-cost shocks	1.201	0.37
FIGARO $K = 64$, het. θ	Single-sector trade-cost shocks	1.101	0.47

Notes: Entries report the WCC/taxonomy welfare-error ratio used in the policy-level diagnostic. Values below one favor welfare-consistent clustering. The win share is the fraction of shocks in the family for which WCC has lower policy-level welfare error; for singleton rows it is mechanically zero or one. The WIOD taxonomy comparator is the broad ISIC-section partition. The FIGARO comparator is canonical FIGARO-64. The “no network/prod. training” row uses 56 USA sector-productivity shocks for evaluation and excludes both network-bundle and productivity shocks from WCC training; policy-level win shares are not reported for that expanded diagnostic.

The large Gulf-proxy ratio in the baseline WCC column illustrates the mechanism: baseline WCC spreads the affected energy sectors across clusters that were not trained for this event, while FIGARO-64 keeps the incidence concentrated in energy groups. Because the Gulf-proxy specification uses FIGARO’s WA rest-of-world source region (which aggregates several Gulf-adjacent economies), this row mechanically broadens the affected-origin set relative to the Saudi-only specification; both are reported in Table 5 and the Saudi-only row is the cleaner reference for the magnitude of the dilution effect. Appendix D reports the cluster-support calculation and the scale-stability checks behind this interpretation.

Together with the official CP2015 NAFTA exercise in Appendix Table E.1, these event diagnostics show the other side of object alignment. The USTR and Brexit examples show taxonomy-friendly incidence maps: even event-specific WCC does not beat FIGARO-64. The energy examples show a different case: taxonomy beats baseline WCC, but event-specific WCC can do better by learning the concentrated energy-incidence geometry directly. The value of WCC is therefore diagnostic. It reveals whether the policy object requires a different response geometry than the inherited classification system, rather than mechanically recommending clustering for every application.

An ex-ante diagnostic for which events are taxonomy-friendly. The conditional structure of these results admits an ex-ante reading: the empirical category of an event (taxonomy-friendly vs. WCC-favorable) should be predictable from observable features of the event-incidence map, independent of the eventual welfare-error ranking. We propose two such predictors and report them across the four event maps in Table 6. The first is the event-support concentration on the inherited taxonomy: the Herfindahl–Hirschman index of FIGARO-64 group shares, where each affected fine

Table 5: Real-event incidence diagnostics: FIGARO-64 errors and relative ratios

Event incidence map	Scale	FIGARO-64 error	Native 73 / FIGARO-64	Baseline WCC / FIGARO-64	Event-oracle WCC / FIGARO-64
USTR Section 301, U.S. on China	Lists 1-3	1.00×10^{-6}	0.513	5.281	1.587
Brexit UK-EU services, ONS exposure	Services-heavy	9.01×10^{-7}	0.837	5.847	2.486
Brexit UK-EU services, ONS exposure	Services-only	6.76×10^{-7}	1.090	7.360	3.430
Russia-EU energy imports	25%	1.28×10^{-5}	0.614	9.762	0.222
Russia-EU energy imports	50%	1.29×10^{-5}	0.634	14.646	0.201
Russia-EU energy imports	100%	1.31×10^{-5}	0.621	18.410	0.212
Hormuz-style energy, Saudi origin	50%	3.18×10^{-6}	0.509	6.513	0.306
Hormuz-style energy, Gulf proxy	50%	1.16×10^{-5}	0.922	35.765	0.263

Notes: The FIGARO-64 error column reports the canonical FIGARO-64 median absolute welfare error. The remaining columns report ratios relative to that error, so values below one indicate lower error than FIGARO-64. Baseline WCC uses the same balance-constrained $K = 64$ labels as the heterogeneous-theta FIGARO comparison. Event-oracle WCC retrain the partition on the event-specific welfare-response geometry; it is forward-looking as an oracle benchmark and retrospective as an implementable partition conditional on a specified incidence map, fine benchmark, and welfare object. The USTR event-oracle row is unconstrained and has a large residual cluster; a cap-12 version gives the same qualitative conclusion, with event-oracle WCC still less accurate than FIGARO-64. Appendix D gives the incidence maps, data sources, commands, and output files.

sector’s baseline trade volume is allocated to its FIGARO-64 parent group and the resulting group totals are normalized to sum to one across the 64 groups (range $[1/64, 1]$, with values closer to one indicating concentration in fewer taxonomy groups). Events with high HHI have incidence localized in a small number of taxonomy buckets and should be approximated well by the taxonomy. The second is the cosine similarity, in fine-sector exposure space, between the event-incidence direction and the top principal component of the between-group covariance of FIGARO-64 group-mean welfare-exposure vectors under the baseline training menu. The saved diagnostics also report the cosine similarity against the top three principal components. Events whose incidence direction lies close to that leading axis are aligned with the variance-preserving direction of taxonomy-induced exposure; events orthogonal to it are not. The joint prediction is that USTR Section 301 and Brexit have high concentration and high cosine similarity (taxonomy-friendly), while the Russia-EU and Hormuz energy events have high concentration but lower cosine similarity to the baseline-menu leading taxonomy axis (the affected sectors are concentrated in two FIGARO-64 energy groups, but their combined incidence direction need not be aligned with the leading taxonomy axis for the baseline policy menu used to train baseline WCC labels).

4.5 Mechanisms and robustness

Table 7 reports the corresponding FIGARO exposure-loss diagnostic for the balance-constrained $K = 64$ heterogeneous-theta labels. The diagnostic is informative, but it should be read as a mechanism diagnostic rather than as a sufficient statistic for welfare error. For targeted bilateral, importer-sector, productivity, and upstream shocks, lower exposure loss lines up with lower welfare error for WCC. For single-sector trade-cost shocks, however, WCC has lower exposure loss but higher welfare error. This mismatch is informative: exposure-space fit diagnoses whether the partition is aligned with the local response target, but it is not mechanically equivalent to counterfactual welfare accuracy once the coarse accounting map, output weights, and finite shocks are imposed. The uniform shock is excluded from the exposure-loss comparison because it has no sector-indexed

Table 6: Ex-ante observable predictors for the four event-incidence diagnostics

Event incidence map	FIGARO-64 HHI	Cosine sim. to taxonomy axis	Empirical category
USTR Section 301, U.S. on China	TBD	TBD	Taxonomy-friendly
Brexit UK–EU services, services-heavy	TBD	TBD	Taxonomy-friendly
Russia–EU energy (50%)	TBD	TBD	WCC-favorable
Hormuz, Gulf proxy (50%)	TBD	TBD	WCC-favorable

Notes: HHI is the Herfindahl–Hirschman index of FIGARO-64 group shares: each affected fine sector’s baseline trade volume is allocated to its FIGARO-64 parent group, and the resulting group totals are normalized to sum to one across the 64 groups; values lie in $[1/64, 1]$, with values closer to one indicating concentration in fewer taxonomy groups. Cosine similarity is computed in fine-sector exposure space between the event-incidence direction (a unit vector over fine FIGARO sectors weighted by event-support trade volumes) and the top principal component of the between-group covariance of FIGARO-64 group-mean welfare-exposure vectors under the baseline training menu used to construct the event-oracle WCC labels in Table 5. The empirical category is read from Table 5: an event is classified taxonomy-friendly if the event-oracle WCC ratio exceeds one, WCC-favorable if it is below one. Both predictors are observable from the event-incidence map and baseline trade volumes alone, without computing any welfare counterfactuals; if they separate the four events ex ante, the conditional structure of Result 2 has an empirically observable diagnostic. *[TBD: Hong populates HHI and cosine-similarity cells from the saved event-runner outputs (coverage_summary.csv and sector_*.rates.csv) combined with the baseline-menu exposure stack saved by the WCC trainer; the latter is the input to the principal-component decomposition.]*

Table 7: FIGARO exposure-loss diagnostic at $K = 64$

Evaluation family	WCC/tax. welfare-error ratio	WCC/tax. exposure-loss ratio
Bilateral full-sector trade-cost	0.126	0.028
Importer-sector trade-cost	0.012	9.0×10^{-25}
Productivity	0.164	0.186
Single-sector trade-cost	1.434	0.179
Upstream-intensive trade-cost	0.346	3.2×10^{-32}
Uniform trade-cost	2.432	—

Notes: Welfare-error ratios use the balance-constrained heterogeneous-theta FIGARO $K = 64$ comparison with canonical FIGARO-64 as the taxonomy benchmark. Exposure-loss ratios compare the within-cluster loss of the WCC partition to canonical FIGARO-64 in the evaluation-family exposure space. Values below one favor WCC. Extremely small exposure-loss ratios arise in low-dimensional targeted exposure spaces where WCC nearly eliminates within-cluster exposure loss. The uniform shock has no sector-indexed exposure target in this diagnostic.

exposure direction in this construction.

Table 8 reports the strongest balance-cap diagnostic at $K = 64$. Tightening the cap does not remove the upstream advantage of welfare-consistent clustering; the upstream WCC/FIGARO-64 ratio falls from 0.376 at cap 40 to 0.265 at cap 12. At the same time, canonical FIGARO-64 remains better for broad uniform shocks, although the WCC penalty narrows under the tightest cap from 2.517 to 1.928. The result is useful because it addresses the most natural concern about WCC: the upstream result is not driven by one residual mega-cluster that simply absorbs hard-to-classify sectors.

Appendix Tables B.5, C.3, and C.4 report additional robustness checks. The shock-size exercise confirms that absolute CP-WIOD errors scale regularly with shock size, while relative performance remains family-specific. The appendix balance-cap tables report the corresponding taxonomy errors and additional shape diagnostics.

Table 8: Balance-cap sensitivity at $K = 64$

WCC cap	Largest cluster	Headline WCC error	Uniform ratio	Upstream ratio
40	40	3.755×10^{-6}	2.517	0.376
30	30	3.695×10^{-6}	2.575	0.362
20	20	4.074×10^{-6}	2.395	0.342
12	12	3.923×10^{-6}	1.928	0.265

Notes: The table reports FIGARO fixed-wage local price-index results at $K = 64$ with partial heterogeneous θ . The headline WCC error is the median absolute welfare error over the full evaluation menu; the corresponding canonical FIGARO-64 headline error is 3.08×10^{-6} . Ratio columns divide WCC error by canonical FIGARO-64 error; values below one favor WCC.

Table 9: CP-WIOD CP2015 full-GE diagnostics at $K = 20$

Diagnostic	All	Uniform	Upstream	Downstream	All wage	All reversal
Target-aligned GE-WCC, baseline shocks	0.588	0.700	0.721	0.483	0.314	−1.06 pp
Fixed baseline GE-WCC labels, $5\times$ shocks	0.574	0.956	1.229	0.309	0.365	−1.54 pp
Downstream holdout GE-WCC, baseline shocks	0.851	0.755	1.028	0.870	0.490	−1.14 pp
Downstream holdout GE-WCC, $5\times$ shocks	0.817	1.235	1.233	0.968	0.517	−1.70 pp

Notes: The table reports WCC/ISIC-taxonomy error ratios from the CP2015 full-GE WIOD exercise with 44 countries and 56 sectors. Welfare columns report median absolute real-income welfare-error ratios by shock family; the “All wage” column reports the WCC/taxonomy ratio for median absolute wage-error; the final column reports the WCC minus taxonomy ranking-reversal frequency in percentage points. Values below one favor WCC in the ratio columns. “Target-aligned” trains WCC on the same GE shock menu used for evaluation. “Downstream holdout” removes the downstream-intensive shock from the WCC training menu but keeps it in the evaluation menu. In this run, the target-aligned GE-WCC labels learned at the baseline shock size are identical to the labels learned at the $5\times$ scale.

5 General-Equilibrium Object Change

The GE diagnostics provide the fourth main empirical lesson: changing the welfare object changes the aggregation target. The baseline exercise remains the fixed-wage local price-index benchmark, which isolates the price-index aggregation margin. The GE exercises instead evaluate real income, $\log \hat{w} - \log \hat{P}$, so aggregation quality also depends on wage incidence and policy rankings.

5.1 CP-WIOD GE diagnostic

The CP-WIOD exercise provides the cleanest GE diagnostic because it can be run through a port of the official [Caliendo and Parro \(2015\)](#) exact-change solver. Table 9 reports this exercise. The key finding is that the apparent weakness of fixed-wage-trained WCC under GE evaluation is partly an objective-mismatch problem. When WCC is trained on GE real-income responses, the all-shock welfare-error ratio falls to 0.588 relative to the ISIC-section taxonomy, and the wage-error ratio falls to 0.314. The downstream holdout remains favorable to WCC, but much less strongly: the downstream welfare-error ratio is 0.870 when the downstream shock is excluded from training. The distinction is between target alignment and out-of-sample generalization. A GE-trained WCC partition is an oracle benchmark for real-income preservation. A holdout exercise instead asks whether response geometry learned from other shocks generalizes to the omitted incidence pattern.

The shock-size experiment adds a useful qualification. Scaling the evaluation shocks by five

does not change the target-aligned GE-WCC labels in the WIOD exercise: the labels learned at the baseline shock size and the labels learned at the $5\times$ scale are identical. Nevertheless, the family-specific performance ratios change. The all-shock ratio remains stable, moving from 0.588 to 0.574, and downstream becomes even more favorable to WCC. But uniform shocks move close to parity and upstream shocks move in favor of the taxonomy. Thus the scale exercise does not indicate unstable clustering. It shows that, even for a fixed partition, larger GE shocks can shift which response margins dominate across shock families.

5.2 NAFTA GE diagnostic

The official CP2015 NAFTA counterfactual, reported in Appendix Table E.1, provides a useful benchmark event because it uses the original 40-sector CP model and the tariff changes from the paper itself. The fine-model real-wage effects from the Python port match the official package: 0.323% for Canada, 1.715% for Mexico, and 0.112% for the United States. The same- K diagnostic shows a case in which the tariff-relevant sector structure is already aligned with the policy incidence: the taxonomy preserves the 20 tradable sectors on which NAFTA tariffs vary, is essentially exact at $K = 21$, and is recovered by NAFTA-oracle WCC. The holdout GE-WCC, trained on generic sector-global GE responses rather than the NAFTA tariff incidence, is much less accurate.

5.3 FIGARO GE pilot with fixed-wage labels

The FIGARO GE diagnostic in Appendix Table E.2 points in the same conceptual direction. The FIGARO replication path uses the same compact exact-change objects as the CP2015 system in Section 2; the paper-oriented runner solves the CP2015 price, expenditure, and wage-residual equations on the compact FIGARO arrays and uses the fixed-trade-imbalance closure when the FIGARO imbalance vector is available. This pilot reuses WCC labels trained for the fixed-wage price-index object and evaluates them under a real-income object. Moving from the fixed-wage local object in (5) to real income shifts the welfare-magnitude comparison toward canonical FIGARO-64; at the same time, welfare-consistent clustering remains competitive on policy rankings and wage-response errors. Thus GE aggregation quality is multidimensional: preserving price-index responses, real-income magnitudes, wage incidence, and policy rankings need not select the same partition. Section 5.4 closes the object-change by retraining WCC on real-income GE response geometry rather than reusing the fixed-wage labels.

5.4 FIGARO GE diagnostic with GE-trained labels

The pilot in Section 5.3 reused fixed-wage WCC labels under a real-income evaluation, combining a change in closure with a change in target. The exercise reported here closes that gap by training WCC directly on real-income GE responses, holding the $K = 64$ compression target and the canonical FIGARO-64 comparator fixed. This is the FIGARO analogue of the CP-WIOD diagnostic in Table 9.

Table 10: FIGARO CP2015 full-GE diagnostics at $K = 64$

Diagnostic	All	Uniform	Upstream	Productivity	Bilateral	All wage
Target-aligned GE-WCC, baseline shocks	TBD	TBD	TBD	TBD	TBD	TBD
Downstream holdout GE-WCC, baseline shocks	TBD	TBD	TBD	TBD	TBD	TBD

Notes: *[Notes to be finalized after results land. Template: WCC error ratios relative to canonical FIGARO-64 under the fixed-trade-imbalance closure with partial heterogeneous- θ ; values below one favor WCC. Reproduction commands and per-row settings will be documented in Appendix D alongside the existing event-incidence runners.]*

Table 10 reports the result. The two diagnostic rows — target-aligned GE-WCC and downstream-holdout GE-WCC — are the FIGARO analogue of Table 9’s headline rows (CP-WIOD all-shock 0.588, wage 0.314).

[Interpretation paragraph: pick one branch from the comment block above, fill numerical placeholders from the runner output, delete the other branches.]

The joint pattern of the pilot in Section 5.3 and the GE-trained exercise reported here is discussed in Section 5.5.

5.5 Interpretation

Sections 5.3 and 5.4 jointly map two GE-relevant aggregation margins. The pilot in Section 5.3 is an object-change diagnostic: it reuses WCC labels trained for the fixed-wage price-index object and evaluates them under a real-income object, so the comparison combines a change in closure with a change in target. Its main lesson is that wage incidence and income effects introduce aggregation margins that are absent from the fixed-wage baseline. The diagnostic in Section 5.4 retrains WCC on real-income GE responses, removing the object-mismatch and reporting how the $K = 64$ FIGARO comparison looks when both the labels and the evaluation share a GE welfare object. Closure-robustness across alternative GE closures and fully economy-wide heterogeneous elasticities remain natural further extensions, but they are distinct from the fixed-wage aggregation exercise that forms the paper’s main evidence.

6 Scope and Practical Guidance

6.1 External validity and the role of welfare-consistent clustering

These results should not be read as proposing welfare-consistent clustering as a universal off-the-shelf replacement for official sector taxonomies. A welfare-consistent partition is benchmark- and objective-dependent: it depends on the available fine data, the policy menu used to construct sector exposures, the closure of the model, and the welfare object used for evaluation. In that sense, what may generalize across datasets is not the exact label assignment itself, but the *procedure* of constructing a partition that preserves the response object of interest.

This distinction matters even when a fine benchmark economy is available. A researcher may still prefer a coarse model for repeated counterfactual grids, uncertainty exercises, transparent reporting,

or comparison with standard WIOD, OECD ICIO, or FIGARO sector systems. In that case, the fine benchmark can be used not as the final reporting object, but as a laboratory for designing and validating the coarse approximation. Welfare-consistent clustering asks what the coarse partition would look like if it were allowed to use the fine economy’s own welfare-response information.

The paper presents welfare-consistent clustering primarily as an *oracle diagnostic at design time*: when an inherited taxonomy and an anticipated policy menu are both fixed, comparing the taxonomy to a baseline-menu WCC partition reveals whether the inherited classification is adequate for the menu, and an event-oracle partition reveals the upper bound on what any aggregation at the same K could achieve. Two further readings of the same evidence are natural practical extensions of this logic rather than separately tested claims. As an *ex-post applied tool*, the event-specific WCC partition is implementable for completed events whose incidence can be specified from observed policy or market information, conditional on the incidence map, fine benchmark, and welfare object; its value is to amortize fine-model response information into a coarse model for repeated post-event exercises, not to provide an out-of-sample forecast. As a *stress-testing* reading, the past event’s oracle partition can serve as a defensible coarse model for sensitivity analysis on a future shock that is expected to share the past event’s incidence structure. We discuss both as practical extensions; neither is separately validated by the present evidence, and both inherit the caveats above: the event-oracle labels depend on which sectors, countries, scales, and welfare object one treats as the event.

Many applications do not have such a benchmark. In those settings, an exact welfare-consistent partition is not identified. The useful lesson is the oracle logic: when a fine economy is available, WCC reveals which response margins a welfare-preserving aggregation would keep and where inherited taxonomies are likely to be inadequate. Without that benchmark, applied work should approximate the same logic with transferable observables or hybrid rules, combining official taxonomies with features such as tradability, input-output position, import exposure, and broad expenditure structure.

6.2 Scope of the evidence

The evidence is designed to answer some questions and to leave others for extensions.

It can identify how taxonomy-like and welfare-consistent aggregations differ under the fixed-wage local price-index object. It can also test whether broad shocks and targeted network-sensitive shocks stress different margins, whether WCC labels generalize across regimes, and whether the results survive exact- K repair, canonical FIGARO-64 comparisons, partial heterogeneous-theta robustness, and balance constraints. Finally, the GE diagnostics show how the comparison changes when the evaluated object becomes real income rather than the fixed-wage price index.

It does not claim that the FIGARO-E3 WCC labels can be exported unchanged to other datasets. The paper now provides GE-trained FIGARO labels at $K = 64$ in Section 5.4, but it does not provide closure-robust GE benchmarks across alternative closures or fully economy-wide heterogeneous elasticities. Those are natural extensions. The contribution here is to recast sector aggregation as a welfare-preserving approximation problem, use WCC as an oracle benchmark when fine data are

available, and show how the object-alignment logic can be studied in a high-resolution global MRIO setting.

6.3 Post-event applicability

The retrospective reading of welfare-consistent clustering is straightforward but worth stating explicitly. After a major event has occurred, the researcher may be able to specify an incidence map from the event itself: which sectors were affected, which origin countries, at what scale, and through which trade channels. The real-world diagnostics in Section 4.4 fit this pattern: the USTR Section 301 tariff lists, the ONS-based UK–EU services map, the post-2022 Russia–EU energy-sanctions environment, and the structural concentration of energy flows through the Strait of Hormuz. The event-oracle partitions in Table 5 are constructed from those incidence maps and the fine benchmark’s response geometry. They are not fitted to the evaluated coarse-model errors. For retrospective coarse-model welfare assessment, the partition is therefore a working aggregation conditional on the chosen incidence map, available fine benchmark, and welfare object, not merely an abstract oracle benchmark.

Three caveats apply. First, ex-post observability is partial. Even after an event, the precise sector boundaries, country composition, and shock magnitude need to be specified, and different specifications produce nearby but not identical partitions. The scale-invariance documented in Section 4.4 is the relevant robustness property: the Russia–EU energy labels are stable from 25% to 100% in our exercise, which means a retrospective user with imperfect scale information can still apply the labels with low penalty. Second, the welfare object remains a modeling choice. The Section 5 GE diagnostics show that the right partition for fixed-wage local price-index welfare need not be the right partition for full-GE real income; an ex-post user must specify the welfare object the partition is meant to preserve. Third, lower welfare error at the same K is a fidelity claim about the coarse approximation, not a claim about the quality of any policy ranking derived from it. In Table 5, event-specific WCC improves on baseline WCC in several rows, but it does not always improve on canonical FIGARO-64.

Within these caveats, the retrospective use of welfare-consistent clustering addresses a practical gap that motivates much of the literature on aggregation. Full-resolution counterfactual analysis is expensive when an event has to be examined under many parameter perturbations, alternative welfare objects, and policy variants. A coarse model with event-specific WCC labels can amortize the fine model’s response information into repeated post-event exercises, especially in cases where the event’s incidence is concentrated in sector groupings that the inherited taxonomy does not preserve. For policy institutions running ex-post welfare attribution of named events, this is the working version of the paper’s diagnostic message.

6.4 Conclusion

The paper’s practical message is simple. Standard taxonomies are useful when policy incidence follows production identity: broad shocks, mild compression, and events defined on official product,

energy, or tariff categories are exactly the cases in which inherited classifications can be the right object. They are more likely to fail when counterfactual incidence runs through welfare-response margins that taxonomy does not preserve, such as targeted network exposure, country-sector incidence, or policy menus that require different sectors to be separated than the official code system would suggest. Two further patterns sharpen this prescription. Compression matters: as the number of retained sectors falls, response-based grouping becomes more valuable because taxonomy-like aggregation discards more welfare-relevant variation. The welfare object also matters: the GE diagnostics in Section 5 show that changing the welfare object can change which partition best preserves the response, so the appropriate aggregation is not invariant to the counterfactual being computed.

Welfare-consistent clustering should therefore be interpreted as an oracle diagnostic rather than a universal replacement for official taxonomies. When a fine benchmark is available, it reveals which welfare-response margins a coarse model would need to preserve and whether an inherited taxonomy is adequate for the policy menu at hand. When a fine benchmark is not available, the lesson is not to export the exact WCC labels mechanically, but to design hybrid or proxy aggregations that combine official sector structure with observable response-relevant features such as tradability, input-output position, import exposure, and final-demand orientation. The relevant aggregation question is not only how many sectors to keep, but which economic object the retained sectors are meant to approximate. For completed events whose incidence can be specified, the same diagnostic logic can also support event-specific coarse partitions, conditional on a fine benchmark and the caveats discussed in Section 6.3.

Appendix

A Data Construction and CP-Compatible Preprocessing

The main text describes the compact benchmark construction used by the solver. The replication workflow starts from the raw FIGARO-E3 product-by-product matrix, parses intermediate-use, final-demand, value-added, and output rows, and converts the raw flows into the compact arrays $(\pi, \gamma, \alpha, \beta, X, VA)$. The cleaning step enforces column-side cost-share consistency, handles zero-output cells with explicit regularization, and constructs a solver-ready bundle used by the fixed-wage, full-GE, and aggregation diagnostics.

Table A.1 reports the audit summary written with the FIGARO-E3 bundle and the corresponding CP-compatible cleaning step. The most important check is column-side consistency. The raw maximum absolute row-column balance discrepancy is small relative to total recorded flows, and the maximum material relative discrepancy is about 0.15%. The solver-side cleaning then clips negative entries and regularizes zero-output country-product cells with $\varepsilon = 10^{-10}$. The clipping amounts look large in levels because the MRIO is recorded in millions of euros, so the table reports scale diagnostics as well: clipped final-demand entries are about 0.123% of positive final demand, clipped value-added entries are about 0.039% of positive value added, and the post-cleaning residual is about 0.054% of positive compact flows. Row balance is not imposed by RAS; the fixed-wage solver uses column-side cost shares, and the remaining row-column residual is reported as a diagnostic.

The largest post-cleaning residual cell is China-CPA.C10.C, with an absolute row-column residual of 18,868.8 mEUR against cleaned column-side output of 33,212.3 mEUR. The next four cells are Germany-CPA.C21, Germany-CPA.M72, GB-CPA.M72, and Germany-CPA.D352.F. These cell-level residuals are reported to make the cleaning audit transparent; they are not used as row-balance restrictions in the fixed-wage price-index solver, which only requires column-side cost-share consistency.

Algorithm A.1: constructing WCC labels. The replication code constructs the labels used in the tables as follows. First, it builds a training menu of shocks and solves the fine benchmark for each shock. The compact FIGARO full-coverage training menu uses sector-global trade-cost shocks for all fine sectors with log size 0.03, productivity shocks for all fine sectors in country index 0 (AR in the standardized FIGARO bundle) with log size 0.03, and upstream-intensive trade-cost bundles with top- k sets $k \in \{5, 20, 50\}$. Upstream scores are the sector’s global intermediate-use share in total use, $\sum_{i,n,r} Z_{inrs} / (\sum_{i,n,r} Z_{inrs} + \sum_n F_{ns})$, and the top- k shocks hit the highest-score sectors. The smaller FIGARO evaluation menu used for mechanism diagnostics samples 30 sector-global trade-cost shocks with seed 0 and log size 0.05, adds a uniform trade-cost shock of log size 0.03, two importer-sector and two bilateral-importer-full-sector shocks of log size 0.10, two productivity shocks of log size 0.05, and two upstream bundles of log size 0.05. The CP-WIOD diagnostic uses the same structure with the parameter values reported in the script metadata.

Second, the code stacks sector-level welfare-exposure vectors by shock family. Each family block

is z-scored using the clustering weights and then multiplied by its scalar family weight; Appendix Table B.3 varies these weights. Third, the code runs weighted K -means on the exposure matrix using global gross-output weights, random seed 0 unless otherwise reported, and 50 initializations. The selected solution is the lowest-inertia K -means solution after the usual weighted objective. If empty clusters remain, exact- K repair repeatedly splits the largest cluster, breaking ties by cluster weight: when taxonomy labels are available it first tries a taxonomy-subgroup split, otherwise it splits the cluster along the weighted first principal component of its within-cluster exposure vectors and assigns the smaller side to a new label. When a balance cap is imposed, the algorithm trims oversized clusters by repeatedly moving the sector that creates the smallest weighted exposure-loss increase into a feasible recipient cluster. Fourth, the resulting labels are not evaluated directly in exposure space: the code aggregates intermediate-use, final-demand, output, value-added, and elasticity objects into a coarse benchmark and then solves the coarse counterfactual model. Random partitions in the main tables are balanced random partitions with the requested K , so their small largest-cluster sizes reflect the balancing rule rather than an unconstrained random draw.

B Additional CP-WIOD Diagnostics

This appendix collects additional diagnostics for the CP-WIOD exercise summarized in Section 3. The purpose is not to make CP-WIOD the paper’s main quantitative setting, but to use a familiar, lower-dimensional Caliendo–Parro-style environment as a diagnostic lab for the aggregation mechanism. Tables B.1–B.5 ask whether welfare-consistent clustering improves aggregation because it is universally superior to sector taxonomies, or because its training objective is aligned with the welfare-response directions being evaluated.

Table B.1 establishes the basic proof of concept: in the canonical 56-sector WIOD benchmark, the ISIC-section taxonomy and WCC are close on pooled welfare errors and both improve on a random partition, while their relative performance differs sharply by shock family. Table B.2 shows that this difference is an alignment result rather than a mechanical network-shock result. Family-oracle WCC is therefore best read as a diagnostic of the evaluation-family exposure space, not as a guaranteed lower bound for every welfare metric. Table B.3 reports objective-weighting sensitivity, Table B.4 makes the resulting clusters interpretable, and Table B.5 reports scaled-shock ratios showing that relative performance remains family-specific away from the baseline shock size.

The appendix is organized to separate three ideas that are easy to conflate in the main text. First, the proof-of-concept comparison asks whether a response-based partition can be competitive with an inherited taxonomy in a canonical quantitative-trade environment. Second, the oracle and holdout comparisons ask whether performance comes from target alignment rather than from a generic advantage of clustering. Third, the weighting and scale exercises ask whether the results are artifacts of arbitrary implementation choices, such as the relative number of shocks in each training family or the baseline shock magnitude. These checks are intentionally compact, but they are useful because the WIOD environment is small enough to make the source of each result transparent.

The main lesson from Appendix B is therefore not that CP-WIOD by itself settles the empirical question. Rather, it provides a controlled laboratory for reading the larger FIGARO results. When the training menu contains response directions similar to the evaluation shock, WCC can improve on taxonomy. When those directions are omitted, WCC can lose its advantage or even underperform. This is the empirical counterpart of Proposition 1: the relevant object is the response geometry spanned by the training menu, not the label “network” or “productivity” by itself.

The proof-of-concept table is deliberately modest. On the pooled all-shock metric, the ISIC taxonomy and WCC are nearly tied. This matters for interpretation: the paper is not built on a claim that official taxonomies are mechanically poor. The informative rows are the family-specific rows. The same two partitions that are close overall behave differently for broad uniform and upstream-intensive shocks, which is the simplest version of the paper’s object-alignment message.

Table B.2 separates three comparison modes. The restricted or holdout column asks whether a partition trained without the relevant response directions still generalizes. The target-aligned column uses the broader training menu from the main CP-WIOD diagnostic. The oracle column trains directly on the evaluation family and should be interpreted as an in-sample exposure-space benchmark. The exposure-loss panel is included because it shows where the clustering objective itself is aligned with the evaluation family before the full coarse counterfactual model is solved.

The weighting sensitivity in Table B.3 addresses a mechanical concern. If one family contributes many more shocks than another, a stacked exposure matrix can overweight that family even before the researcher makes an economic choice about which margins matter. The baseline therefore normalizes exposure blocks by family and then applies scalar family weights. The table shows that the target-alignment pattern is not coming from one knife-edge set of scalars. Removing block normalization is the most demanding stress test, and it mainly weakens the pooled all-shock comparison rather than overturning the network and productivity-family patterns.

Table B.4 makes the CP-WIOD labels less black-box. Some clusters are close to conventional production categories, while others group sectors with similar exposure to intermediate demand or final demand even when their ISIC codes differ. This is exactly the kind of object-specific regrouping that WCC is designed to detect. The table is not meant to provide a universal taxonomy; it illustrates how a response-based partition can differ from an inherited production-code partition in economically interpretable ways.

Finally, Table B.5 holds the baseline WCC labels fixed while scaling the evaluation shocks. This is not a re-optimization exercise. It asks whether a local response-based partition remains informative away from the neighborhood in which it was trained. The answer is mixed in an informative way. Absolute errors scale regularly, which is reassuring for the solver and metric, but the WCC/taxonomy ratio is not invariant to shock size. Large shocks can activate response margins that are weak at the baseline scale, and locally trained labels can become less aligned with the finite-shock counterfactual.

Table A.1: FIGARO-E3 parsing and CP-compatible cleaning audit

Audit item	Value
<i>Panel A. Raw parsing and ETL output</i>	
Raw matrix data rows	9,809
Raw matrix data columns	10,028
Countries	46
Fine products	213
Native suffix-collapsed groups	73
Country-product rows / columns	9,798 / 9,798
Value-added rows	11
Final-demand columns	230
Ignored rows / ignored columns	0 / 0
Nonzero intermediate-use flows	11,658,469
Nonzero final-demand flows	157,218
Maximum absolute row-column discrepancy	0.272 mEUR
Maximum material relative discrepancy	0.154%
Absolute discrepancy relative to total flows	2.06×10^{-9}
Elasticity input in saved bundle	Uniform $\theta_s = 4$
<i>Panel B. CP-compatible cleaning</i>	
Intermediate-use negative cells clipped	165; 576.243 mEUR (0.00086% of positive intermediate use)
Final-demand negative cells clipped	7,025; 80,020.359 mEUR (0.123% of positive final demand)
Value-added negative cells clipped	79; 25,538.646 mEUR (0.039% of positive value added)
Zero-output country-product cells regularized	1,759
Post-cleaning row-column residual	max 18,868.793 mEUR; total 106,878.987 mEUR (0.054% of positive compact flows)
Country-product cells in compact benchmark	9,798

Notes: Panel A summarizes `data/figaro_e3_2015_bundle/etl_summary.csv` and the raw FIGARO-E3 matrix header. Panel B is derived from the saved parquet bundle using the cleaning rules in `src/preprocess.py`. Negative entries are clipped to zero. Zero-output cells are regularized by setting column-side output and value added to $\varepsilon = 10^{-10}$. The preprocessing enforces column-side cost-share consistency but does not RAS-balance row totals.

Table B.1: Canonical CP-WIOD proof-of-concept

Aggregation	Shock family	Median abs. error	Spearman ρ	Reversal freq.
ISIC-section taxonomy	All shocks	0.000020	0.9938	0.0226
Welfare-consistent	All shocks	0.000019	0.9930	0.0228
Random	All shocks	0.000034	0.9909	0.0278
ISIC-section taxonomy	Uniform trade-cost	0.000166	—	—
Welfare-consistent	Uniform trade-cost	0.000200	—	—
ISIC-section taxonomy	Upstream-intensive	0.000344	—	—
Welfare-consistent	Upstream-intensive	0.000168	—	—

Notes: The table reports fixed-wage local price-index welfare errors relative to the 56-sector WIOD benchmark. The elasticity vector maps the tradable-sector estimates in [Caliendo and Parro \(2015\)](#) to WIOD and assigns a fallback value to services and nontradables. The taxonomy comparator groups sectors by broad ISIC section, producing $K = 20$ groups. Ranking statistics are omitted for singleton shock-family rows.

Table B.2: CP-WIOD oracle and exposure-loss diagnostics at $K = 20$

Evaluation family	Welfare-error ratio			Exposure-loss ratio		
	Restricted/holdout	Target-aligned	Family-oracle	Restricted/holdout	Target-aligned	Family-oracle
Single-sector trade cost	0.958	0.957	0.994	0.111	0.255	0.111
Upstream bundle	0.970	0.452	1.029	0.978	0.000	0.000
Downstream bundle	0.624	0.195	0.392	0.652	0.000	0.000
Productivity	1.050	0.956	0.826	0.354	0.204	0.057

Notes: Welfare-error entries are WCC/taxonomy median welfare-error ratios. Exposure-loss entries are within-cluster exposure losses relative to the taxonomy in the same evaluation-family exposure space. Values below one favor WCC. “Restricted/holdout” uses the WCC trained without network and productivity bundle directions; for single-sector trade-cost shocks, this restricted menu still contains single-sector trade-cost shocks and is therefore not a strict holdout. “Oracle” trains WCC only on the evaluation family. Uniform trade-cost shocks are excluded because they have no sector-indexed exposure target in this construction. The five-seed median is reported for restricted/holdout and target-aligned WCC; oracle rows use the baseline seed.

Table B.3: CP-WIOD objective-weighting sensitivity at $K = 20$

Training objective	All	Uniform	Upstream	Downstream	Productivity
Baseline block-z-score weights	0.943	0.712	0.452	0.195	0.956
Equal family weights	0.818	0.646	0.432	0.160	0.829
Network-heavy weights	0.943	0.712	0.452	0.195	0.956
Productivity-heavy weights	0.769	0.573	0.380	0.150	0.708
No block z-score	1.009	0.680	0.480	0.209	0.908

Notes: Entries are WCC/ISIC-taxonomy welfare-error ratios for the target-aligned CP-WIOD diagnostic. Values below one favor WCC. The baseline uses family-level weighted z-scores and scalar weights $(\lambda_\tau, \lambda_{\text{network}}, \lambda_{\text{targeted}}, \lambda_{\text{prod}}) = (1, 1, 0.5, 0.5)$. Equal family weights set all four scalars to one. Network-heavy sets $\lambda_{\text{network}} = 2$; productivity-heavy sets $\lambda_{\text{prod}} = 2$. The final row turns off family-level block z-scoring and is included as a stress test. The qualitative target-alignment result does not rely on one arbitrary scalar-weight choice, although removing block normalization weakens the pooled all-shock result.

Table B.4: Interpreting representative CP-WIOD WCC clusters

Representative WCC cluster	Size	Output share	Tax. purity	Upstreamness	Final-demand share
C13–C15; J61; C27; J62–J63	18	12.8%	0.386	0.604	0.396
G47; I; P85; R–S	4	9.9%	0.318	0.258	0.742
C23; C22; H52; C17	9	5.7%	0.681	0.886	0.114
C20; M69–M70	2	4.8%	0.543	0.869	0.131
C24; C25	2	4.3%	1.000	0.918	0.082

Notes: The table reports selected clusters from the representative seed-zero target-aligned CP-WIOD WCC partition at $K = 20$. Sector codes follow the WIOD industry classification. Output share is the cluster share of global gross output. Taxonomy purity is the output share of the dominant ISIC-section label inside the WCC cluster. Upstreamness and final-demand share are output-weighted cluster averages; they summarize the share of sector sales absorbed by intermediate users and final demand, respectively.

Table B.5: CP-WIOD shock-size sensitivity

Shock family and diagnostic	Baseline	2×	5×	10×
<i>Panel A. WCC/taxonomy welfare-error ratios</i>				
All shocks, fixed baseline WCC labels	0.919	0.904	0.913	0.909
Single-sector trade-cost, fixed baseline WCC labels	0.919	0.905	0.913	0.919
Uniform trade-cost, fixed baseline WCC labels	1.203	0.992	0.807	0.718
Upstream-intensive, fixed baseline WCC labels	0.489	0.582	0.857	0.951
Downstream-intensive holdout	0.736	0.826	0.764	0.785
Single-productivity holdout	0.311	0.341	0.351	0.311
<i>Panel B. Single-productivity holdout normalized error levels</i>				
Taxonomy relative error	20.04%	18.40%	13.28%	17.30%
WCC relative error	6.24%	6.28%	4.66%	5.38%
WCC/taxonomy ratio	0.311	0.341	0.351	0.311

Notes: Panel A entries are WCC/taxonomy median absolute welfare-error ratios in the CP-WIOD exercise. Values below one favor welfare-consistent clustering. Panel B reports productivity-holdout errors normalized by the fine-model welfare response. “Fixed baseline WCC labels” holds the WCC partition learned at the baseline shock size fixed while scaling the evaluation shocks. “Holdout” rows exclude the indicated family from the WCC training menu.

C FIGARO Robustness and Mechanisms

This appendix collects the FIGARO diagnostics that support the mechanism and robustness claims in the main text. Table C.1 shows that the $K = 64$ welfare-consistent clusters are economically interpretable but need not coincide with canonical FIGARO-64 groups: some clusters preserve familiar product identities, while others deliberately cross taxonomy boundaries to group sectors with similar welfare-response profiles. Table C.2 separates the role of input-output propagation from regime-specific relabeling by comparing the regime used to train WCC labels with the regime used to evaluate them.

Tables C.3 and C.4 address the concern that WCC results are driven by an oversized residual cluster. The balance-constrained specifications reduce the largest WCC cluster, and the stronger cap grid tightens the restriction further at $K = 64$. Across these restrictions, the main qualitative pattern survives: canonical FIGARO-64 remains stronger for broad uniform shocks under mild compression, whereas WCC retains a large advantage for upstream-intensive shocks. The appendix therefore supports the interpretation that the relevant margin is target alignment and propagation, not a clustering artifact.

The FIGARO appendix has a different role from the CP-WIOD appendix. CP-WIOD is used as a transparent diagnostic lab; FIGARO is the high-resolution environment in which the paper’s main empirical claims are evaluated. The checks below therefore focus on concerns that are specific to a high-dimensional MRIO setting. Do the WCC labels have an economic interpretation? Are the results driven by IO propagation or by relabeling under a particular regime? Are the results an artifact of a giant residual cluster? The tables answer these questions by looking directly at cluster composition, cross-regime evaluation, and balance-cap sensitivity.

The main point is not that every robustness table favors WCC. In fact, some of the appendix evidence reinforces the strength of taxonomy for broad shocks and mild compression. The stronger claim is that the central shock-family contrast survives the natural implementation objections: taxonomy preserves broad production identity well, while WCC remains useful for targeted upstream incidence even when cluster size is constrained and labels are evaluated across IO regimes.

The examples in Table C.1 show why WCC should not be read as an alternative statistical classification of products. Singleton clusters such as construction and real estate remain intact because their welfare-response profiles are distinctive and large. Other clusters combine products from different FIGARO-64 groups when they have similar positions in the response geometry. The low-purity clusters are therefore not automatically failures. They are the places where the response object asks for a grouping that differs from production identity.

Table C.2 helps interpret the no-IO exercise in the main text. Turning off input-output linkages sharply reduces aggregation errors, especially for WCC, which suggests that network propagation is an important amplification channel for structural aggregation error. The cross-evaluation rows also show that regime-specific relabeling is not the main source of the result. Labels trained with IO and evaluated without IO, or trained without IO and evaluated with IO, behave in a way that is consistent with the target-alignment interpretation: the evaluation regime matters because it

changes the response object being approximated. The extremely small no-IO WCC errors arise because the no-IO local price-index object is close to sector-separable in the exposure space used for clustering; once network propagation is removed, the response-based partition can almost exactly preserve the remaining local exposure directions.

Table C.3 reports the first balance intervention. The purpose is not to force WCC to mimic taxonomy, but to check whether the unconstrained clustering routine is obtaining fit by creating one large residual cluster and many small special-purpose clusters. The answer is reassuring but nuanced. Balance constraints improve the shape of the partition and do not erase the shock-family pattern, although they can move the headline all-shock and uniform-shock rows. This is why the paper treats cluster balance as a robustness and presentation concern rather than as a separate economic object.

The stronger cap grid in Table C.4 makes the same point more directly. Tightening the maximum cluster size from 40 to 12 does not make WCC resemble the canonical taxonomy in the uniform shock, and it does not eliminate the upstream advantage. If anything, the upstream ratio improves as the cap becomes tighter. This pattern is useful for interpretation because it rules out the simplest mechanical critique of WCC: the upstream result is not merely a by-product of hiding many sectors inside a residual mega-cluster.

Table C.1: Interpreting representative FIGARO WCC clusters at $K = 64$

Representative WCC cluster	Size	Output	Purity	Upstream	Final
CPA_F_A	1	6.8%	1.000	0.183	0.817
CPA_L	1	6.5%	1.000	0.268	0.732
CPA_H53; CPA_N79; CPA_S95; ...	54	5.4%	0.184	0.683	0.317
CPA_C24_E; CPA_C24_N; CPA_B08_A; ...	12	0.8%	0.701	0.992	0.008
CPA_R90T92; CPA_S96; CPA_D3511_A; ...	44	4.0%	0.248	0.419	0.581

Notes: The table reports selected clusters from the balance-constrained heterogeneous-theta FIGARO WCC partition at $K = 64$. Output share is the cluster share of global gross output. FIGARO-64 purity is the output share of the dominant canonical FIGARO-64 group inside the WCC cluster. Upstreamness and final-demand share are output-weighted cluster averages.

Table C.2: Cross-evaluation across IO regimes at $K = 64$

Labels regime	Evaluation regime	WCC median $ \Delta \log W $	Comparator
With IO	With IO	3.73×10^{-6}	Canonical FIGARO-64: 3.08×10^{-6}
No IO	With IO	3.95×10^{-6}	Canonical FIGARO-64: 3.08×10^{-6}
With IO	No IO	1.59×10^{-9}	No-IO FIGARO-64: 2.85×10^{-7}
No IO	No IO	1.97×10^{-10}	No-IO FIGARO-64: 2.85×10^{-7}

Notes: The table reports all-shock median welfare errors for balance-constrained WCC labels in the partial heterogeneous-theta $K = 64$ FIGARO specification. “Labels regime” indicates whether WCC exposures are computed with input-output linkages or under the no-IO ablation; “evaluation regime” indicates the model used to evaluate the resulting labels. The comparator column reports the corresponding taxonomy error in the same evaluation regime.

Table C.3: Balance-constrained welfare-consistent clustering: shape versus fit

Specification	K	Max cluster	All	Uniform	Upstream
Unbalanced WCC, het. θ	64	91	3.99×10^{-6}	7.68×10^{-5}	6.21×10^{-6}
Balanced WCC, het. θ	64	54	3.73×10^{-6}	1.03×10^{-4}	7.98×10^{-6}
Unbalanced WCC, het. θ	73	85	2.17×10^{-6}	5.57×10^{-5}	5.60×10^{-6}
Balanced WCC, het. θ	73	54	2.27×10^{-6}	7.73×10^{-5}	6.14×10^{-6}

Table C.4: Stronger balance-cap sensitivity at $K = 64$

WCC cap	Largest cluster	All WCC error	All ratio	Uniform ratio	Upstream ratio
40	40	3.76×10^{-6}	1.217	2.517	0.376
30	30	3.70×10^{-6}	1.198	2.575	0.362
20	20	4.07×10^{-6}	1.321	2.395	0.342
12	12	3.92×10^{-6}	1.272	1.928	0.265

Notes: The table reports the FIGARO fixed-wage local price-index comparison at $K = 64$ with partial heterogeneous θ . The taxonomy benchmark is exact canonical FIGARO-64. “All WCC error” is the median absolute welfare error of the WCC partition over the evaluation menu. Ratio columns divide the WCC median welfare error by the corresponding canonical FIGARO-64 error; values below one favor WCC.

D Event-Incidence Details

This appendix gives the event-incidence recipes behind Section 4.4. It is intentionally written partly as a replication appendix; readers interested only in the economic incidence maps can focus on Tables D.1–D.4 and the short energy-event construction paragraph. Each event is used as an incidence map, not as a claim about realized historical welfare effects. The common workflow is the same across events. We first specify affected origins, destinations, sectors, and diagnostic ad-valorem changes. We then solve the fine FIGARO model, evaluate inherited taxonomies and baseline WCC labels, and optionally retrain event-oracle WCC labels on the event-specific response geometry. The output files have the same structure whenever possible: a partition summary, a country-level welfare-error file, a sector-rate file, and a coverage file documenting the event support.

D.1 USTR Section 301 China tariff lists

The Section 301 exercise treats the official U.S. tariff lists as an observed product-list incidence map. It is not an estimate of the realized welfare effects of the 2018–2019 U.S.–China trade war. It maps listed HTS products into FIGARO sectors and asks which sector partition best preserves the fine FIGARO welfare response to the implied U.S. import-cost shock on China.

Table D.1 establishes that, for this event, inherited product taxonomies are already closely aligned with the policy incidence. The native 73-group taxonomy-like partition and canonical FIGARO-64 both produce smaller welfare-magnitude errors than the baseline $K = 64$ WCC partition. Event-oracle WCC improves on baseline WCC, but it remains an in-sample oracle diagnostic and does not beat FIGARO-64 in this event. This supports the paper’s object-alignment interpretation: event-specific product-list shocks can be cases where official sector definitions are the right aggregation object.

The mapping proceeds in three steps. The USTR lists are first read at the HTS line level and collapsed to HS6 codes. Those HS6 products are then mapped into CN2015 and CPA2.1 categories using the same bridge used in the heterogeneous-theta exercise. Finally, the affected CPA2.1 sectors are matched to fine FIGARO products and weighted by baseline U.S. imports from China. The resulting shock is therefore not a stylized uniform tariff increase. It is a product-list incidence map, with tariff exposure concentrated in the sectors covered by the official policy lists.

This construction is useful precisely because it is not designed to favor WCC. The Section 301 lists are written in official product language, so a taxonomy based on product identity has a natural advantage. The event-oracle WCC row should therefore be read as an upper-bound diagnostic: it asks how much a response-based partition could improve if it were allowed to train directly on this event’s incidence map. The fact that this oracle improves on baseline WCC but does not dominate FIGARO-64 reinforces the point made in Section 4.4: when policy incidence is already organized around official product categories, inherited taxonomies can be the right object.

The ranking statistics in Table D.1 also illustrate why the paper reports more than one metric. Baseline WCC preserves the cross-country ranking well for this event, but it performs poorly on

welfare magnitudes because the affected product-list incidence is diluted across response-based clusters that were not trained for this tariff event. FIGARO-64 and the native taxonomy-like partition make smaller magnitude errors because they keep the official product incidence closer to its original support. This distinction between magnitude preservation and ranking preservation appears again in the GE diagnostics.

The saved Section 301 runs are produced by `scripts/run_figaro_ustr301_event.py`. The script reads the cached USTR JSON file in `data/raw/ustr_section301/`, downloads it if absent unless `--no-download` is set, maps HTS rows to HS6 and then to the CN2015–CPA2.1 bridge, and writes the standard event summary, coverage, sector-rate, mapping, and country-welfare CSVs. The main event-oracle row can be reproduced with:

```
python3 scripts/run_figaro_ustr301_event.py \
  --include-event-oracle-wcc \
  --output results/figaro_ustr301_event_oracle
```

D.2 Brexit UK–EU services

The Brexit services diagnostic uses the same event-incidence logic but with services rather than goods. We start from the ONS time-series table for UK trade in services by service type and partner country and take 2019 “Total EU27” exports plus imports as the service-exposure measure (Office for National Statistics, 2026). Table D.2 reports the mapping from ONS service categories into FIGARO service sectors. The rates are diagnostic ad-valorem equivalents, not estimates of post-Brexit trade barriers. They scale a 10 percent services baseline by the square root of each category’s ONS exposure share, with rates clipped between 4 and 16 percent. This gives larger shocks to transport, finance, professional and management consulting, technical and other business services, and travel-related services, while still keeping the exercise transparent and reproducible.

The ONS workbook used by the runner is stored in `data/raw/ons/` under the filename `uk_trade_services_type_by_partner_current.xlsx`. The parser uses sheet `Sheet 1. Time Series`, header row 2, country code B5 (Total EU27), and year 2019. The file checksum in the saved run is

6b9ebac94f7a45f0a75feb334c93ce4cad94cdfa000f1a73f8766926ca7fa23d.

For ONS category c , the diagnostic service rate is

$$\tau_c - 1 = \text{clip} \left[0.10 \left(\frac{s_c}{\bar{s}} \right)^{1/2}, 0.04, 0.16 \right],$$

where s_c is the category’s share of UK–EU services and $\bar{s}$ is the mean positive category share. The square-root scaling keeps the diagnostic heterogeneous without allowing the largest ONS categories to dominate the experiment mechanically.

The two main Brexit rows in Table 5 are reproduced by:

```
python3 scripts/run_figaro_brexit_event.py \
  --service-rate-mode ons_exposure --variant services_heavy \
  --scenario hard --include-event-oracle-wcc --oracle-max-cluster-size 12 \
```

Table D.1: Real-world tariff-incidence diagnostic: USTR Section 301 China lists

Partition	Largest cluster	Median $ \Delta \log W $	Relative error	Spearman ρ	Ratio to FIGARO-64
Native taxonomy-like 73	22	5.15×10^{-7}	1.39%	0.991	0.513
Canonical FIGARO-64	22	1.00×10^{-6}	2.72%	0.990	1.000
Baseline WCC $K = 64$	54	5.30×10^{-6}	14.36%	0.997	5.281
Event-oracle WCC $K = 64$	145	1.59×10^{-6}	4.32%	0.988	1.587

Notes: The table evaluates a U.S. import-cost shock on China using USTR Section 301 Lists 1–3. HTS entries are mapped to HS6 and then to FIGARO sectors through the CN2015–CPA2.1 bridge used in the heterogeneous-theta mapping. The diagnostic uses 7,369 HTS rows, of which 5,133 map to the FIGARO bridge, covering 2,596 HS6 codes, 32 FIGARO official goods sectors, and 148 fine FIGARO sectors. The covered sectors account for 93.6% of baseline U.S. imports from China in the FIGARO benchmark, with a trade-weighted additional tariff rate of 23.4%. “Relative error” divides the median absolute aggregation error by the median absolute fine-model welfare response across countries. The event-oracle WCC is trained on the Section 301 incidence map; it improves on the baseline WCC but is deliberately not interpreted as an out-of-sample result.

Table D.2: ONS service exposure used in the Brexit UK–EU diagnostic

Script category	ONS service category	Share of UK–EU services	Diagnostic rate	FIGARO sector stems
transport	Transportation	13.5%	12.9%	H49–H53
travel_hospitality	Travel / hospitality	23.9%	16.0%	I, N79
construction	Construction	1.5%	4.2%	F
insurance_pension	Insurance and pension	2.2%	5.1%	K65
financial	Financial	12.2%	12.2%	K64, K66
intellectual_property	Intellectual property	5.5%	8.3%	J58, J59–60
ict	ICT	7.0%	9.3%	J61, J62–63
rd	Research and development	3.2%	6.2%	M72
professional_management	Professional and management	12.4%	12.3%	M69–70, M73
technical_other_business	Technical and other business	13.8%	13.0%	M71, M74–75, N77, N78, N80–82
personal_cultural	Personal and cultural	1.8%	4.7%	R90–92, R93
government	Government	0.9%	4.0%	O84

Notes: Shares use ONS 2019 UK trade in services with Total EU27, summing exports and imports. The mapped categories cover 97.8% of total UK–EU services in the ONS table. Diagnostic rates scale a 10 percent services baseline by the square root of category exposure and are clipped to the interval [4%, 16%]. The services-heavy specification in Table 5 applies these service rates plus a 2 percent goods friction; the services-only specification applies only the service rates.

```
--output results/figaro_brexit_event_services_ons_heavy_hard_oracle_cap12

python3 scripts/run_figaro_brexit_event.py \
  --service-rate-mode ons_exposure --variant services_only \
  --scenario hard --include-event-oracle-wcc --oracle-max-cluster-size 12 \
  --output results/figaro_brexit_event_services_ons_only_hard_oracle_cap12
```

The services-heavy row affects 156 fine sectors and 57 official sectors; the affected support covers 75.9 percent of UK–EU bilateral trade in the FIGARO benchmark and has a trade-weighted diagnostic rate of 6.26 percent. The services-only row affects 30 fine sectors and 26 official sectors; the affected support covers 48.7 percent of bilateral trade and has a trade-weighted diagnostic rate of 5.72 percent. The ONS service categories mapped in Table D.2 cover 97.8 percent of the ONS UK–EU services total.

The Brexit result is useful because it was designed to be a harder services case. Services are often exactly where official taxonomies look coarse, and the ONS exposure map makes the shock heterogeneous across finance, professional services, ICT, transport, and related categories. Yet the event remains taxonomy-friendly in Table 5. Event-oracle WCC improves substantially on the baseline WCC partition, but it does not beat canonical FIGARO-64. The lesson is not that services aggregation never matters. It is that service heterogeneity by itself is not enough to favor WCC when the incidence map is already close to official service categories.

Table D.4 reports a category-holdout version of the same diagnostic. The evaluation shock is the full ONS services-only Brexit map, but the WCC training menu is restricted to a subset of ONS service categories. This is a stricter test than the event-oracle row because it asks whether response geometry learned from one part of the services incidence map transfers to the remaining categories. It does not in this exercise. Training on all services gives an event-oracle ratio of 3.43 relative to canonical FIGARO-64. Excluding professional, technical, and R&D services raises the ratio to 8.83; training only on finance, insurance, ICT, intellectual property, and transport raises it to 11.51; training only on professional, technical, and R&D services gives 11.43. These holdout results reinforce the object-alignment interpretation. The Brexit services map is not a setting where a generic response-based partition uncovers a robust taxonomy failure. The incidence geometry is category-specific, and the official service taxonomy remains the better coarse approximation in this benchmark.

The three holdout rows are reproduced by:

```
python3 scripts/run_figaro_brexit_event.py \
  --service-rate-mode ons_exposure --variant services_only \
  --include-event-oracle-wcc --oracle-max-cluster-size 12 \
  --oracle-exclude-categories professional_management,technical_other_business,rd \
  --output results/figaro_brexit_event_services_ons_holdout_no_proftech_cap12

python3 scripts/run_figaro_brexit_event.py \
  --service-rate-mode ons_exposure --variant services_only \
  --include-event-oracle-wcc --oracle-max-cluster-size 12 \
  --oracle-train-categories transport,financial,insurance_pension,ict,intellectual_property \
```

```

--output results/figaro_brexit_event_services_ons_holdout_fin_ict_transport_cap12

python3 scripts/run_figaro_brexit_event.py \
--service-rate-mode ons_exposure --variant services_only \
--include-event-oracle-wcc --oracle-max-cluster-size 12 \
--oracle-train-categories professional_management,technical_other_business,rd \
--output results/figaro_brexit_event_services_ons_holdout_proftech_cap12

```

These category-holdout runs change only the WCC training menu. The evaluation shock remains the full ONS services-only Brexit map in every row.

D.3 Russia–EU and Hormuz-style energy events

The two energy diagnostics use the same runner structure as the Section 301 and Brexit exercises, but the incidence maps are much simpler. The Russia–EU diagnostic raises EU27 import costs on Russian energy sectors. The affected FIGARO stems are CPA_B05, CPA_B06, CPA_C191, and CPA_C192, corresponding to coal, crude oil and gas, coke, and refined petroleum. The Hormuz-style diagnostic raises foreign import costs on Gulf-origin oil, gas, and refined energy exports. Its affected stems are CPA_B06 and CPA_C192. The Saudi-origin specification uses exporter SA; the Gulf-proxy specification uses SA,WA, where WA is FIGARO’s rest-of-world source region containing several economies. This is why the Gulf-proxy row is labeled illustrative in Table 5.

The energy rows in Table 5 are reproduced by:

```

python3 scripts/run_figaro_russia_energy_event.py \
--ad-valorem 0.25 --include-event-oracle-wcc --oracle-max-cluster-size 12 \
--output results/figaro_russia_energy_event_eu27_x25_oracle

python3 scripts/run_figaro_russia_energy_event.py \
--ad-valorem 0.50 --include-event-oracle-wcc --oracle-max-cluster-size 12 \
--output results/figaro_russia_energy_event_eu27_x50_oracle

python3 scripts/run_figaro_russia_energy_event.py \
--ad-valorem 1.00 --include-event-oracle-wcc --oracle-max-cluster-size 12 \
--output results/figaro_russia_energy_event_eu27_x100_oracle

python3 scripts/run_figaro_hormuz_event.py \
--exporters SA --ad-valorem 0.50 \
--include-event-oracle-wcc --oracle-max-cluster-size 12 \
--output results/figaro_hormuz_event_sa_x50_oracle

python3 scripts/run_figaro_hormuz_event.py \
--exporters SA,WA --ad-valorem 0.50 \
--include-event-oracle-wcc --oracle-max-cluster-size 12 \
--output results/figaro_hormuz_event_gulfproxy_x50_oracle

```

Each output directory contains `summary.csv`, `coverage_summary.csv`, `sector_tariff_rates.csv`, and `welfare_by_country.csv`. The `coverage_summary.csv` files record the exporter/importer sets,

energy stems, affected fine and official sectors, and trade-weighted diagnostic rate. The event-oracle labels are trained on single-sector versions of the same event incidence map, so they should be read as event-specific response partitions rather than generic energy-shock labels.

The Gulf-proxy row in Table 5 is the clearest example of incidence dilution. The shock hits 22 fine energy sectors, all in crude oil and refined energy. Canonical FIGARO-64 keeps this incidence concentrated in two energy groups: the affected sectors account for about 92 percent of the support in the oil-and-gas group and 98 percent in refined energy. Baseline WCC spreads the same affected sectors across seven clusters. In the largest such cluster, only 8 of 54 sectors are affected and the event support share is about 9 percent, so the coarse shock is averaged with many unrelated sectors. This is why the WCC absolute error is large, not merely why the FIGARO-64 denominator is small: the median baseline-WCC error is 4.14×10^{-4} , compared with 1.16×10^{-5} for canonical FIGARO-64. Event-oracle WCC reverses this by learning the energy-incidence geometry directly, with a largest cluster of six sectors in the saved Gulf-proxy run.

The Russia–EU rows also provide a useful scale check. The event-oracle ratios are nearly invariant to shock size, moving from 0.222 at 25 percent to 0.201 at 50 percent and 0.212 at 100 percent. In this exercise, the event-specific labels encode the response geometry of the affected energy sectors rather than the size of the perturbation. That is the robustness property that matters for retrospective use: once the affected sector set is fixed, imperfect knowledge of the exact shock magnitude is less damaging than using a generic partition that was not trained on the event support.

Table D.3: Brexit runner output manifest

Output file	Contents
<code>summary.csv</code>	Partition-level errors, rank statistics, cluster sizes, and ratios to FIGARO-64
<code>coverage_summary.csv</code>	Event support, rates, source/destination sets, ONS mapping metadata, and coverage shares
<code>sector_brexit_rates.csv</code>	Fine-sector diagnostic rates and ONS service-category assignments
<code>ons_service_category_exposure.csv</code>	ONS category values, shares, FIGARO stems, and implied service rates
<code>welfare_by_country.csv</code>	Country-level fine and coarse welfare responses for each partition
<code>run_manifest.json</code>	Script arguments, output list, source checksum, and coverage metadata

Notes: The manifest is written by `scripts/run_figaro_brexit_event.py`. It records the exact command-line arguments, output directory, ONS workbook checksum, core input-file checksums, Python and package versions, and the current git commit when available.

Table D.4: Brexit ONS services category-holdout diagnostic

WCC training menu	Training fine sectors	Median error	Ratio to FIGARO-64
All ONS service categories	30	2.32×10^{-6}	3.430
All except professional / technical / R&D	22	5.97×10^{-6}	8.829
Finance / insurance / ICT / IP / transport only	15	7.78×10^{-6}	11.510
Professional / technical / R&D only	8	7.72×10^{-6}	11.427

Notes: The evaluation shock is the ONS services-only UK–EU Brexit diagnostic in Table 5. The taxonomy benchmark is canonical FIGARO-64, whose median error is 6.76×10^{-7} . The first row is the in-sample event-oracle WCC row from Table 5. The remaining rows train WCC labels on only part of the ONS services incidence map and evaluate on the full services-only map. Values above one indicate larger error than FIGARO-64.

E General-Equilibrium Diagnostics

This appendix reports the general-equilibrium diagnostics that support Section 5. The point is not to replace the fixed-wage local benchmark with a final full-GE ranking of aggregation rules. Instead, the appendix shows how the aggregation target changes when the welfare object changes from price-index preservation to real-income preservation. Table E.1 uses the official CP2015 NAFTA counterfactual as a same- K event diagnostic, while Table E.2 reports the larger FIGARO GE pilot.

The two tables make complementary points. The NAFTA exercise is a case where the policy incidence is defined on the original tariffed tradable sectors, so a tariff-relevant taxonomy is nearly exact and NAFTA-oracle WCC recovers the same structure. The FIGARO GE pilot is different: it reuses fixed-wage-trained WCC labels under a real-income welfare object, so the comparison should be read as an object-change diagnostic rather than a final ranking of aggregation rules. Together, the results reinforce the paper’s main message that the right aggregation depends on the welfare object and policy menu being approximated.

The GE appendix is intentionally more cautious than the fixed-wage analysis. In the fixed-wage benchmark, the aggregation target is the local cost-of-living response in the price block. In the GE diagnostics, welfare is real income: both prices and wages move, and the coarse partition must preserve incidence through the expenditure and labor-market closure as well as through production-network prices. A partition trained for the fixed-wage object need not be the right partition for the GE object. For that reason, the tables below are interpreted as object-change diagnostics rather than as a claim that either taxonomy or WCC is uniformly preferred in GE.

The NAFTA grid is produced by `scripts/run_cp2015_nafta_k_grid.py` with output directory `results/cp2015_nafta_k_grid`. The saved metadata records the official CP2015 NAFTA-only tariff counterfactual, $K = \{10, 15, 20, 21, 25, 30\}$, 50 WCC initializations, harmonic theta pooling, and convergence of the fine 40-sector benchmark. The compact command is:

```
python3 scripts/run_cp2015_nafta_k_grid.py \
--output results/cp2015_nafta_k_grid
```

The FIGARO GE pilot is produced by `scripts/run_figaro_ge_pilot.py` and saved in `results/figaro_ge_pilot_k64`. Its output files are `decomposition.csv`, `welfare_by_policy.csv`, and `labels.csv`. The table below uses the same GE closure described in Section 2: wages clear the expenditure residual under the chosen closure, and welfare is real income rather than the fixed-wage price-index object.

The NAFTA table is included because it is anchored in the official CP2015 replication environment. It is a useful limiting case: when the policy shock is itself defined on the original tariffed tradable sectors, a tariff-relevant taxonomy can be nearly exact at the same K . In such cases, a response-based oracle does not reveal a failure of taxonomy; it recovers the same incidence structure. The holdout GE-WCC column is the relevant generalization test, and it shows that labels trained on generic GE trade-cost responses do not automatically transfer to the NAFTA tariff-incidence object.

The ratio version of Table E.1 is intentionally not reported. For the tariff-relevant taxonomy rows, the denominator is nearly zero because the comparator is designed around the same tariff support

as the policy shock. Ratios would therefore be visually dramatic but economically uninformative. Reporting absolute errors makes the lesson clearer: the taxonomy and NAFTA-oracle WCC are both close to exact, while holdout GE-WCC is not.

Table [E.2](#) extends the object-change message to the FIGARO environment. Canonical FIGARO-64 is more accurate on median GE welfare magnitudes in this narrow pilot, while WCC has a lower ranking-reversal rate and a lower median wage error. The table therefore should not be summarized as “taxonomy wins” or “WCC wins.” It shows that GE aggregation quality is multidimensional. A partition that preserves real-income magnitudes need not be the same partition that best preserves wage incidence or policy rankings. This is why the paper keeps the fixed-wage local benchmark as the clean baseline object and uses GE to show how the aggregation target changes when the welfare object changes.

Table E.1: Official CP2015 NAFTA same- K full-GE diagnostic

Taxonomy comparator	K	Taxonomy error	NAFTA-oracle WCC error	Holdout GE-WCC error
Ordered frontier	10	1.92×10^{-5}	1.92×10^{-5}	2.57×10^{-5}
Ordered frontier	15	1.72×10^{-5}	1.04×10^{-5}	2.46×10^{-5}
Ordered frontier	20	1.08×10^{-5}	9.70×10^{-7}	1.16×10^{-5}
Ordered frontier	21	1.51×10^{-5}	1.08×10^{-7}	1.19×10^{-5}
Ordered frontier	25	1.44×10^{-5}	2.78×10^{-7}	1.44×10^{-5}
Ordered frontier	30	1.88×10^{-5}	3.03×10^{-7}	1.17×10^{-5}
Tariff-relevant taxonomy	21	1.08×10^{-7}	1.08×10^{-7}	1.19×10^{-5}
Tariff-relevant taxonomy	25	2.07×10^{-7}	2.78×10^{-7}	1.44×10^{-5}
Tariff-relevant taxonomy	30	1.27×10^{-7}	3.03×10^{-7}	1.17×10^{-5}

Notes: The table uses the official CP2015 40-sector NAFTA-only tariff counterfactual. Entries are median absolute real-income welfare errors relative to the fine 40-sector benchmark. The ordered frontier uses contiguous sector blocks in the original CP ordering. The tariff-relevant taxonomy keeps the 20 tariffed tradable sectors separate and aggregates the remaining nontradables. NAFTA-oracle WCC is trained on sector-specific NAFTA tariff-response vectors. Holdout GE-WCC is trained on generic sector-global GE trade-cost responses and excludes the NAFTA tariff-incidence object. Absolute errors are reported because the tariff-relevant taxonomy is nearly exact, making WCC/taxonomy ratios unstable and economically uninformative.

Table E.2: FIGARO full-GE diagnostic at $K = 64$

Aggregation	Largest cluster	Median $ \Delta \log W $	Reversal rate	Median $ \Delta \log w $
Canonical FIGARO-64	22	1.46×10^{-4}	4.61%	1.43×10^{-3}
Welfare-consistent clustering	54	2.43×10^{-4}	3.42%	1.17×10^{-3}

Notes: The table evaluates a narrow FIGARO full-GE menu at $K = 64$. Welfare is real income, so the welfare error combines wage and price-index responses. The WCC labels are the fixed-wage labels reused under the GE object, not a fully GE-trained FIGARO partition. The table is therefore an object-change diagnostic: it shows how different GE margins can favor different partitions.

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